Modern businesses depend on Cloud computing to operate and scale their services effectively in the digital age. Cloud Solution Architecture must be understood in order to fully leverage the benefits of Cloud computing. Identifying the core architectural components of the various Cloud service models, including Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS), is part of this process.

Moreover, it is important to outline the Cloud delivery model, which determines how services are delivered to end users, and the Cloud deployment model, which describes how Cloud resources are distributed across various geographical locations and its different types of controlling mechanisms. Through an illustration of the Cloud deployment model, businesses can gain a better understanding of how they can optimize their Cloud infrastructure to achieve high availability, scalability, and minimize costs associated with Cloud computing. It is essential that businesses have a clear understanding of Cloud Solution Architecture in order to obtain the maximum benefit from Cloud Computing and gain a competitive edge in their respective markets.

This first module will cover the basic understanding of cloud solution architecture and its various core components. The following concepts will be covered:

1. Fundamentals of Cloud Solution Architecture
2. Various Cloud service models – Iaas, Paas, Saas
3. Core architectural components
4. Cloud delivery model; and
5. Cloud deployment model

As an excellent introduction into this course and this module, please watch the following video titled “Cloud computing in 6 Minutes”.

VIDEO LINK(HTTPS://WWW.YOUTUBE.COM/WATCH?V=M988_FSOSWO)

Learning Objectives

By the end of this module, you should be able to complete the following:

01 Recognize key pillars of cloud solution architecture.

02 Identify benefits of cloud architecture.

03 Illustrate Cloud service and deployment models.

Key Terms and Concepts

Key Term or Concept	Definition
SaaS	Software as a Service
PaaS	Platform as a Service
IaaS	Infrastructure as a Service

Initial Readings

Although not required, it is recommended that you read chapter 4 of the following text:

Thomas Erl_ Zaigham Mahmood_ Ricardo Puttini - Cloud Computing_ Concepts, Technology & Architecture-Prentice Hall (2013) -Chapter 4

In this module, you learned foundational knowledge regarding Cloud Solution Architecture, more specifically, its core components, benefits of cloud, cloud adoption framework and cloud service and deployment models. In the next module, you will learn to design and implement various cloud compute services and take deep dive into a reference architecture for a concrete use case.

References

Simplilearn. 2021. Cloud Computing In 6 Minutes | What is cloud computer? | Cloud computing explained | Simplilearn (YouTube) Video. Retrieved from **https://www.youtube.com/watch?v=M988_fsOSWo (https://www.youtube.com/watch?v=M988_fsOSWo)**

Thomas Erl_ Zaigham Mahmood_ Ricardo Puttini - Cloud Computing_ Concepts, Technology & Architecture-Prentice Hall (2013) -Chapter 4

Cloud deployment model

A cloud deployment model defines the cloud services consumed and the responsibility model managed by it. It also outlines the relationship between the cloud infrastructure and the users. To determine which model could be suitable for the organisation, we need to understand all four models first and identify a variety of use cases in which these might be the best fit. The four Cloud deployment models include:

1. Public Cloud
2. Private Cloud
3. Hybrid Cloud
4. Multi Cloud

Let's take a look at each of these in greater detail.

Public Cloud

A public cloud deployment does not own any hardware or infrastructure. All the resources are provided by a cloud service provider. The biggest advantage of the cloud services are elastic scalability, resource efficiency, and reducing upfront capital investment. It's optimized for performance, but this is less cost-effective. AWS, Azure, IBM cloud and Google Cloud are all popular public cloud providers.

Advantages

One of the major advantages of using public cloud services is that they are built to support workloads of virtually any scale, from small start-ups to large enterprises. The computing resources provided by cloud providers are vast, including servers, storage, and networking equipment that can be scaled up or down depending on the needs of a particular workload.

Another advantage of using public cloud services is the ease of provisioning without upfront cost. Pay-as-you-go pricing is typically provided by cloud providers, so that organizations only pay for the resources that they use on an hourly or monthly basis. In this way, provisioning infrastructure is free of upfront costs, and organizations can easily add or remove resources as needed.

Additionally, public cloud providers offer a massive catalog of services that can be used to build and run applications in the cloud. These services include storage, databases, computing, networking, security, and more. Utilizing these services allows organizations to develop complex applications without having to manage the underlying infrastructure.

Disadvantages

One of the main disadvantages of using public cloud services is shared tenancy. Due to the shared nature of cloud resources, there is a risk of noisy neighbors, where the performance of one customer's workload may affect the performance of another customer's workload. There are, however, measures built into cloud providers to mitigate this risk.

Another disadvantage of using public cloud services is that data residency and privacy regulations might not be satisfied in the public cloud. In some organizations, data residency requirements dictate where data can be stored and processed, and the public cloud may not be able to meet those requirements. Furthermore, some organizations may have concerns about the privacy of their data when it is stored and processed in the public cloud.

Furthermore, another disadvantage of using public cloud services is that organizations pay for everything they use. If resources are not adequately managed, pay-as-you-go pricing can lead to unexpected costs for some workloads. Cloud service usage and costs must be closely monitored by organizations to avoid overspending.

Private Cloud

The private cloud is a dedicated environment for a single enterprise or organization. Hardware and resources are not shared outside your company. There is a possibility of getting a private cloud from a public cloud provider. Resources are dedicated to a single organization. Some well-known private cloud providers include: Red Hat Open Stack, Rackspace, Microsoft Azure Stack and VMware Private Cloud.

Advantages

One of the major advantages of using a private cloud is that there is less risk of exposing sensitive data, as the tenancy is limited to your organization. You will have greater control over the security of your data and can put in place more stringent security measures. You are also able to implement more granular access controls and monitoring because you control more of the physical and logical infrastructure.
Another advantage of using a private cloud is that you have greater control over data storage residency and data privacy concerns. Organizations may require their data to be stored and processed in a particular location, and a private cloud can provide more flexibility in this respect.

Furthermore, private clouds offer much more restrictive access, increasing security. As access to the cloud resources is restricted to your organization, there is less risk of unauthorized access or data breaches.

Disadvantages

One of the main disadvantages of using a private cloud is that the catalog of service offerings is considerably more restrictive than public cloud offerings. Because the private cloud is usually constructed using dedicated infrastructure, the resources are not shared between multiple customers. Therefore, the range of services available is limited.

Another disadvantage of using a private cloud is that pricing is considerably higher than public cloud services. A private cloud is a dedicated environment for a single organization, so the cost of building and maintaining that environment is entirely funded by the organization.

Hybrid Cloud

Hybrid Cloud is the combination of a public and private cloud. Creating a hybrid cloud means that an organization is using the public cloud but also owns on-premises systems. Due to security requirements or data protection, some companies can't operate only in the public cloud, so they choose the hybrid cloud.

Advantages

One of the major advantages of hybrid cloud solutions is that they enable organizations to migrate from on-premises infrastructure to scalable cloud solutions. It also allows organizations to gradually move workloads to the cloud, thereby minimizing the risk and cost associated with cloud migration.

Hybrid cloud reduces the pain and cost in cloud migration while you run out the cost of your data center. Hybrid cloud solutions enable organizations to keep their most sensitive data on-premise and gain greater control over privacy concerns, while still utilizing cloud services for non-sensitive workloads.

Disadvantages

One of the main disadvantages of hybrid cloud solutions is that infrastructure management is complex and challenging. It can be time-consuming and require specialized skills for organizations to manage both on-premises and cloud infrastructure. In addition, hybrid cloud solutions may have compatibility issues due to the integration of on-premises infrastructure and cloud services.

Another disadvantage of hybrid cloud solutions is that migrating between on-premises and cloud services is not always a one-to-one mapping. On-premises infrastructures and cloud services may handle data and applications differently, which can create challenges during migration.

Scalability of on-premises systems is still challenging and can cap the scalability of your system. Hardware limitations restrict the scalability of on-premises infrastructures. It is likely that organizations will have to invest in additional hardware in order to meet the demand, which could be costly.

Finally, outages may be harder to diagnose and debug as the cloud architecture is more complex. Hybrid cloud solutions include on-premises infrastructure as well as cloud services, which can contribute to outages. Organizations will have a more difficult time diagnosing and debugging outages, resulting in longer downtime and increased costs for them.

Multi Cloud

Multi cloud model uses more than one cloud provider at a time, where you use both the public and private cloud. In multi-cloud, instead of combining private with public, you'd use more than one public cloud. The motivations for a multi-cloud deployment are for redundancy or to optimize cost for preferred technology choices. The most common reason for deploying multi-cloud is when you need a specific service from one public cloud provider and another service from another public cloud provider.

Advantages

One of the main advantages of multi-cloud solutions is that they can reduce vendor stickiness and lock-in. Multiple cloud providers enable organizations to reduce their dependence on a single provider and avoid being locked into a single vendor.

Another advantage of multi-cloud solutions is that they can potentially increase redundancy depending on the cloud architecture.

Additionally, multi-cloud solutions reduce limits in the catalog of cloud solutions available to organizations. Organizations are able to access a broader range of services and can choose the best solution according to their requirements. It is possible to increase redundancy and reduce the risk of downtime by using multiple cloud service providers. For mission-critical workloads, this can enhance reliability and availability.

Disadvantages

One of the main disadvantages of multi-cloud solutions is that moving data in the cloud costs you as you consume cloud service providers' bandwidth.

Another disadvantage of multi-cloud solutions is that they require the cultivation of multiple deployment/pipeline strategies for delivering to multiple cloud service providers. In order to manage this effectively, specialized skills are required.

Considerably more complex cloud architecture leads to difficult-to-observe and troubleshoot systems. Diagnostics and ensuring optimal system performance can be challenging when there are multiple cloud service providers and interdependencies between them.

Finally, it is very challenging to optimize and find cost savings in multi-cloud solutions because of the distribution of costs and resources. There can be difficulties in tracking and managing costs when using multiple cloud service providers, and resources may be underutilized or overprovisioned. As a result, costs may increase, and efficiency may be reduced.

Cloud service models

A cloud delivery model represents a specific, pre-packaged combination of IT resources offered by a cloud
provider. There are three major models of cloud architecture that are driving organizations to the cloud.

1. Infrastructure-as-a-Service (IaaS)
2. Platform-as-a-Service (PaaS)
3. Software-as-a-Service (SaaS)

Over the course of the next few pages, we will examine each of these driving organizations a little further. Let's get started with IaaS.

Infrastructure as a Service (IaaS)

IaaS is essentially the process of renting a data center from a cloud provider. IaaS is not the service model of choice to accomplish a digital transformation, but there are couple of scenarios that we can tackle with IaaS. For instance, the lift-and-shift of existing workloads to the cloud can be achieved through IaaS. The purpose of this approach is to relocate applications and data from on-premises systems to the cloud in order to take advantage of the scalability and flexibility of cloud computing infrastructure. This is often accomplished without significant modifications.

IaaS is a good alternative for smaller companies that can't invest in their own data center. These companies can focus on their core business activities by leveraging the services provided by a cloud service provider without worrying about maintaining an on-premises infrastructure.

In the context of a disaster recovery strategy, when adding a cloud-based data center to your existing on-premises servers ensures business continuity during a disaster by offering a backup infrastructure which can be activated rapidly. To speed up the time to market, some legacy practices and processes were adjusted upfront to align with the cloud delivery model. A cloud-native application can be adapted from an existing application, or new cloud-native applications can be developed and optimized to work on the cloud infrastructure. In this way, organizations can accelerate their time-to-market.

Platform as a Service (PaaS)

PaaS is a fully managed service model that helps you build new solutions (or refactor existing ones) much faster. Unlike IaaS, in order to make a successful cloud journey, PaaS requires a heavy involvement from all the layers of the company. With PaaS, much of the infrastructure and most operations are delegated to the cloud provider. The multi-tenant offerings are cost-friendly, and you can easily leverage the economies of scale, providing you adopt the PaaS model. PaaS is

suitable for many scenarios, such as the modernization of existing workloads, for example the company can migrate its legacy systems to the cloud and incorporate machine learning and data analytics into its system by using tools and frameworks provided by the PaaS platform.

Software as a Service (SaaS)

SaaS is a fully managed business suite of software that often relies on a cloud platform for its underlying infrastructure. This model hosts and manages end-user applications. Here, there’s no need to deploy applications and software locally. They are delivered and maintained over the internet and usually need a web interface to be accessed. SaaS examples include Salesforce and Adobe Creative Cloud, as well as Microsoft's own Office 365, Power BI, and Dynamics 365

Benefits

By providing a range of benefits previously unavailable or only available to large enterprises, the cloud computing architecture has revolutionized the way businesses operate. The benefits of adopting a cloud computing architecture include, but are not limited to:

Elimination of Capital Expenditure

- The elimination of the capital expenditure involved in using an on premise server, storage, and networking infrastructure. This advantage is that, businesses are able to access a range of IT resources that would otherwise be prohibitively expensive, thus reducing their overall IT expenses and enhancing their profitability.

Shift IT resources to the Public Cloud

- Cloud architecture allows organizations to shift their IT resources to the public cloud, which provides them with greater flexibility, agility, and scalability. Businesses can therefore adapt quickly to market conditions and customer demands, thereby improving their competitiveness.

Solves Latency Issues and Improves Data Processing

- Cloud solves latency issues and improves data processing requirements enabling businesses to quickly access and process large amounts of data. Using real-time data, businesses are able to make more informed decisions and improve overall productivity and efficiency.

Easily Scale Up and Down Resources

- Cloud helps businesses to easily scale up scale down cloud resources based on their changing needs. In this way, businesses are able to focus on their core business activities instead of making large capital expenditures on IT infrastructure.

Encourages Remote Working and Promotes Collaboration

- Cloud encourages remote working and promotes team collaboration by providing employees with access to a variety of cloud-based services and tools. The flexibility of working remotely increases the productivity and efficiency of employees.

Automatic Updates

- Cloud automatically updates its services eliminating the need for businesses to perform manual updates. Therefore, businesses will have access to the latest technologies and software.

Recovery and Security

- It results in better disaster recovery and provides high security allowing businesses to continue to access their critical data and applications in the event of a catastrophe. In addition to providing security protocols for protecting their clients' data, cloud providers also have their own security protocols.

Competitive Advantage

- It gives businesses a competitive advantage. Generally speaking, cloud architecture provides businesses with a range of benefits. These include a decrease in IT expenditures, an increase in operational efficiency, and an increased level of competitive advantage.

Cloud Adoption Framework

The Microsoft Cloud Adoption Framework for Azure is a full lifecycle framework that enables cloud architects, IT professionals, and business decision makers to achieve their cloud adoption goals. It provides best practices, documentation, and tools that help you create and implement business and technology strategies for the cloud.

Take this opportunity to explore the following resource, as it will expand on the Cloud Adoption Framework by providing an overview, scenarios, as well as, tools and templates to help get you started.

For proven guidance on each phase of the cloud journey and best practices that help you confidently adopt the cloud and achieve business outcomes, please visit the following resource for additional information.

MICROSOFT CLOUD ADOPTION FRAMEWORK FOR AZURE (HTTPS://LEARN.MICROSOFT.COM/EN-US/AZURE/CLOUD-ADOPTION-FRAMEWORK/)

Core Components

Cloud architecture is the way technology components combine to build a cloud, in which resources are pooled through virtualization technology and shared across a network. In general, the components of cloud architecture can be categorized into several components that work together seamlessly to provide cloud services and to create an infrastructure for running applications and managing data. The components of a cloud architecture include:

- A front-end platform (the client or device used to access the cloud)
- A back-end platform (servers and storage)
- A cloud-based delivery model; and
- A network

Let’s take a deeper look at these components.

Front-end Platform

Cloud architecture begins with the front-end platform, which is the client or device that accesses the cloud. A graphical user interface (GUI) or command-line interface (CLI) is provided by this platform to enable users to interact with the cloud. There are many devices that can be connected to the front-end platform, including laptops, smartphones, tablets, and other Internet of Things (IoT) devices.

A back-end platform

In the cloud, these components are responsible for hosting applications, data, and other resources that end-users have access to. Backend platform components include several layers of hardware and software infrastructure, including virtualization software and various networking protocols.

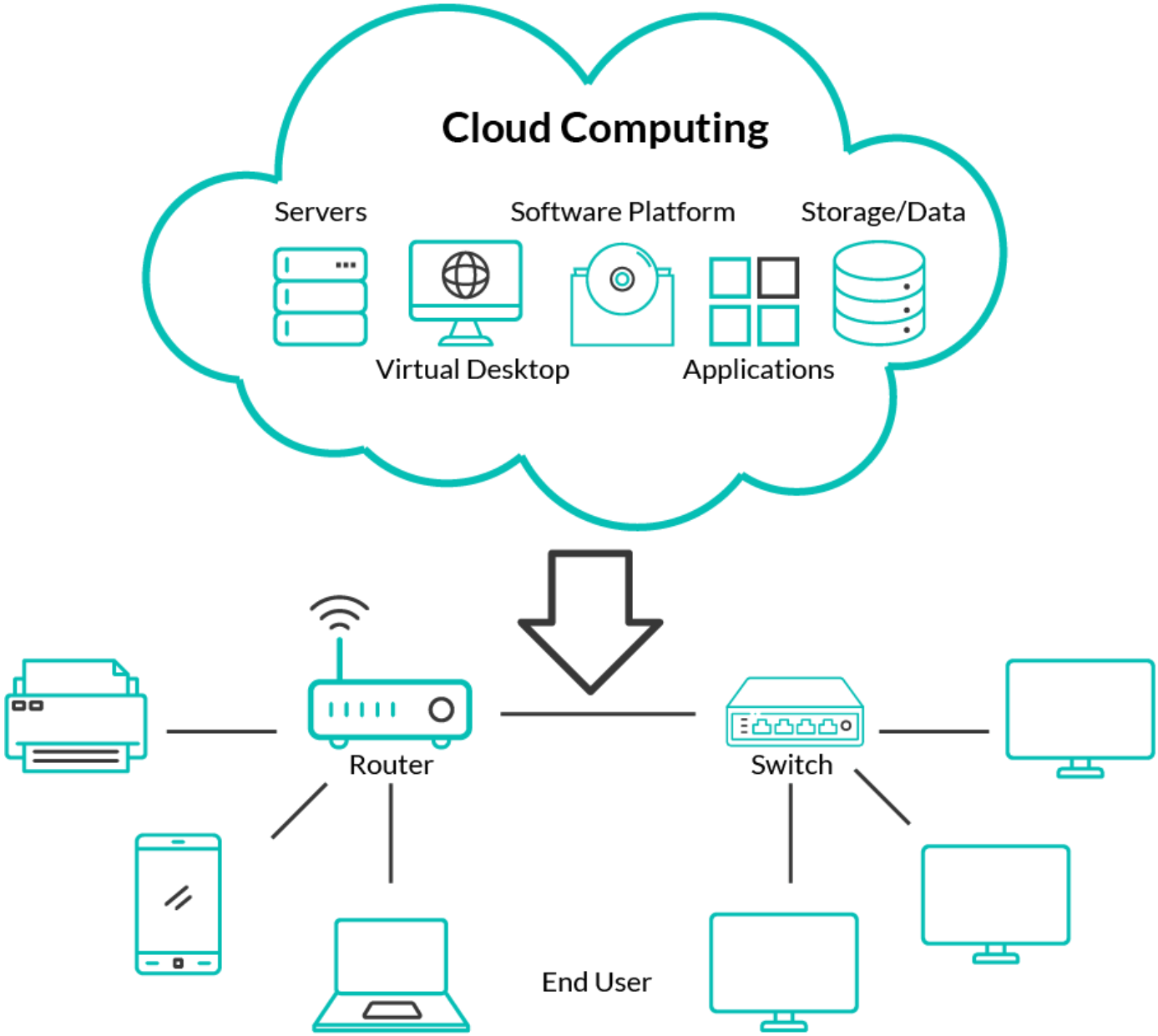
A cloud-based delivery model

Cloud services are delivered over the internet, thereby eliminating the need for on-premises infrastructure. A web browser or application installed on a user's device can be used to access cloud services from anywhere in the world.

Network

In cloud architecture, the network component plays an important role in enabling communication between the various components. Cloud infrastructure includes a variety of networks such as local area networks (LANs), wide area networks (WANs), and the Internet, as well as networking devices such as routers, switches, and firewalls that ensure secure communication between the various components.

As you can see in the diagram below, together, these technologies create a cloud computing architecture on which applications can run, providing end-users with the ability to leverage the power of cloud resources.



Additionally, to create a cloud solution architecture, the following must be present:

1. **Client Infrastructure:**

This is the front-end component that helps when interacting with the cloud. In order for users to interact with the cloud, the front-end platform, which can be a device or application that is used to access cloud-based services, is a critical component of cloud architecture.

2. **Application:**

This is the software piece that clients want to access.

3. **Service:**

The service component manages which type of service you can access according to the client's requirements. Cloud computing offers three types of services, including:

a. Software as a Service (SaaS)
b. Platform as a Service (PaaS)
c. Infrastructure as a Service (IaaS).

These types will be described in greater detail later in the module.

4. Runtime Cloud:

This offers execution and runtime environment to VM.

5. Storage:

This component provides large amount of storage capacity in cloud to store and manage data.

6. Infrastructure:

It offers services on the host level, network level, and application level. Cloud infrastructure includes hardware and software components like servers, storage, network devices, and virtualization software.

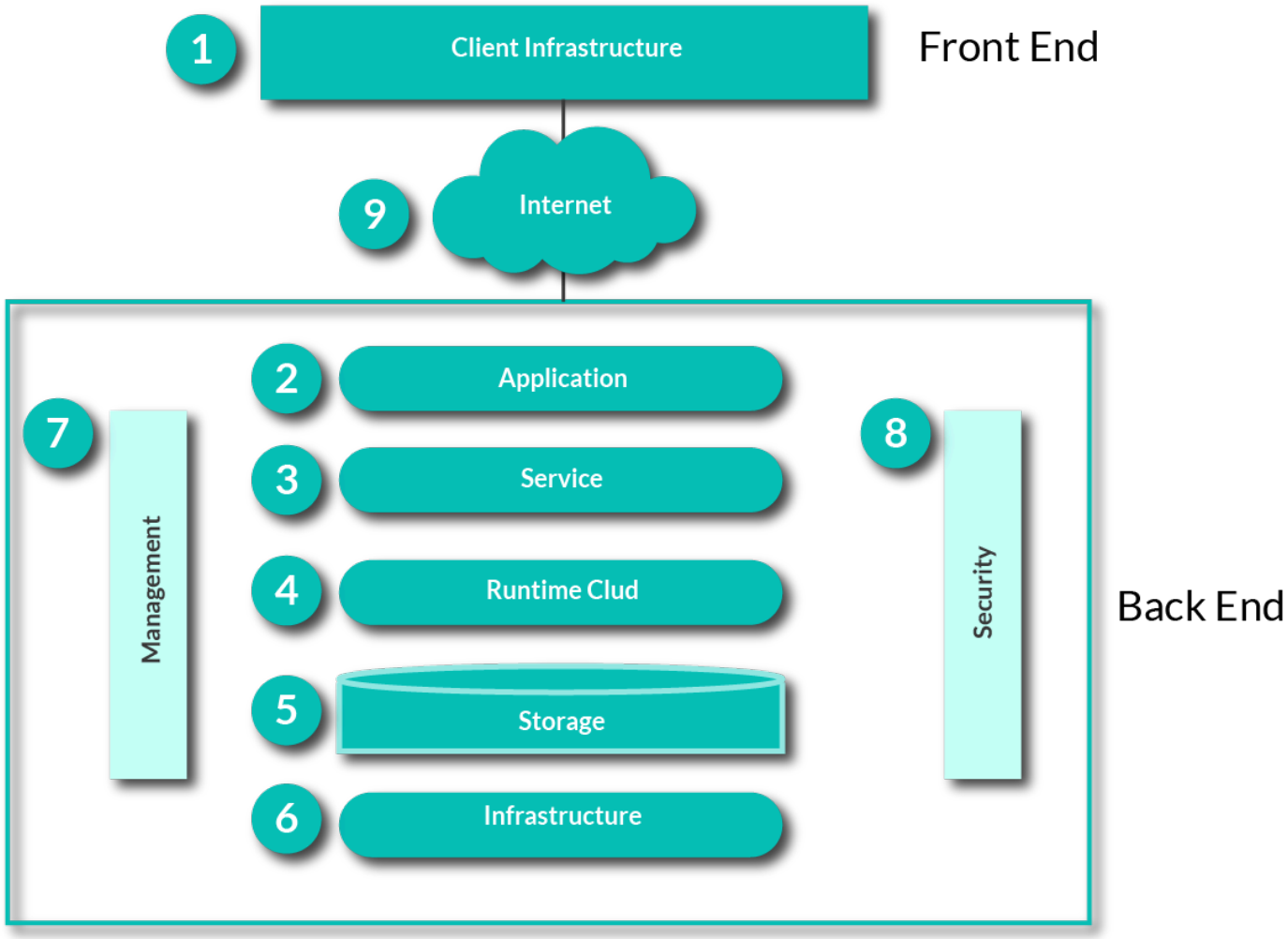
7. Management:

This component manages components like application, service, runtime cloud, storage, infrastructure, and other security matters in the backend. It also establishes coordination between them.

8. Security:

Security in the backend refers to implementing different security mechanisms for secure Cloud systems, resources, files, and infrastructure to the end-user.

How these components are connected to create a cloud solution architecture is depicted in the image below.


1
Client Infrastructure
Front End
9
Internet
2
Application
7
Management
3
Service
8
Security
4
Runtime Clud
5
Storage
6
Infrastructure
Back End

As a technology, cloud computing has revolutionized the way we access and use technology resources. Using cloud computing solutions, you will be able to access scalable, on-demand, and cost-effective computing resources, including computing power, storage, and network capabilities.

Cloud computing design refers to the process of developing cloud computing solutions that meet the specific needs of an organization or business. In order to design a cloud solution, you must identify the requirements, select the appropriate cloud service provider and services, design the architecture of the solution, and implement it in a scalable, reliable, and secure manner. This process ensures that cloud solutions are designed and implemented in a way that meets business requirements and provides the desired benefits of cloud computing.

In this module, we will describe and design cloud computing solutions for various Azure services. The following concepts will be covered:

1. Design for Azure virtual machines solutions
2. Design for Virtual Machine scale sets
3. Design for Container Instances
4. Design for Azure Kubernetes service solutions
5. Design for App Service; and
6. Design for azure function solutions

Learning Objectives

By the end of this module, you should be able to complete the following:

01 Illustrate Azure Virtual Machines

02 Interpret Virtual Machine scale sets

03 Clarify Azure Container Instances

04 Explain Azure Kubernetes Service

05 Recall Azure App Service

06 Identify Azure Functions

Key Terms and Concepts

Key Term or Concept	Definition
AKS	Azure Kubernetes Service

In this module, you learned various compute services provided by Azure and how to provision resources in Azure. A number of Azure services are described in some detail, such as Azure Virtual Machines, Virtual Machine scale sets, Azure Container Instances, Azure Kubernetes Service, Azure App Service, and Azure Functions.

In the next module, you will learn to design and implement various cloud networking services and deep dive into a reference architecture for a concrete use case.

References

Azure Developers. 2017. Azure Functions Overview (YouTube) Video. Retrieved from **https://youtu.be/8-jz5f_JyEQ (https://youtu.be/8-jz5f_JyEQ)**

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Microsoft Mechanics. 2021. Increase app availability and scalability | Azure Virtual Machine Scale Sets (YouTube) Video. Retrieved from **https://www.youtube.com/watch?v=lE2xJXYHnB8 (https://www.youtube.com/watch?v=lE2xJXYHnB8)**

Design Principles

Microsoft has identified 10 high-level design principles that will make your application more scalable, resilient, and manageable. These design principles apply to any architectural style. Throughout the design process, keep these 10 high-level design principles in mind.

- **Design for self-healing**. In a distributed system, failures happen. Design your application to be self-healing when failures occur.
- **Make all things redundant**. Build redundancy into your application, to avoid having single points of failure.
- **Minimize coordination**. Minimize coordination between application services to achieve scalability.
- **Design to scale out**. Design your application so that it can scale horizontally, adding or removing new instances as demand requires.
- **Partition around limits**. Use partitioning to work around database, network, and compute limits.
- **Design for operations**. Design your application so that the operations team has the tools they need.
- **Use managed services**. When possible, use platform as a service (PaaS) rather than infrastructure as a service (IaaS).
- **Use an identity service**. Use an identity as a service (IDaaS) platform instead of building or operating your own.
- **Use the best data store for the job**. Pick the storage technology that is the best fit for your data and how it will be used.
- **Design for evolution.** All successful applications change over time. An evolutionary design is key for continuous innovation.
- **Build for the needs of business**. Every design decision must be justified by a business requirement.

Reference: Microsoft, 2022.

Learning Activity 2.1: Reference architecture [Ungraded]



As a final element to this module, let's follow a scenario and define a reference architecture for it to help solidify your understanding of Cloud design.

This example will show you how Azure services such as virtual machine scale sets, Virtual Network, Key Vault, Storage, Load Balancer, and Monitor can be quickly provisioned for the deployment of an efficient private Corda network where member banks can establish their own nodes. This scenario covers the back-end components that are necessary to create a scalable, secure, and monitored private, enterprise distributed ledger technology (DLT) network within a consortium of two or more members.

Scenario: This example scenario is useful for banks or any other institutions that want to establish a trusted environment for information sharing without resorting to a centralized database. In this example, we will describe the scenario in the context of maintaining credit score information between banks, but the architecture can be applied to any scenario where a consortium of organizations wants to share information with one another without resorting to the use of a central system ran by one single party.

Traditionally, banks within a financial system rely on centralized sources, such as credit bureaus, for information on an individual's credit score and history. A centralized approach presents a concentration of operational risk and sometimes an unnecessary third party.

Dataflow

1. Bank A creates/updates an individual's credit record by creating a transaction.
2. Data flows from Bank A's private application server to the **Azure Load Balancer (https://learn.microsoft.com/en-us/azure/load-balancer/)**, and then to a node VM on the virtual machine scale set.
3. A transaction proposal is created on the network ledger.
4. The transaction is committed to the ledger, when all the required signatures are gathered.
5. Bank B can read the credit record created by bank A by communicating with its own node.

Components

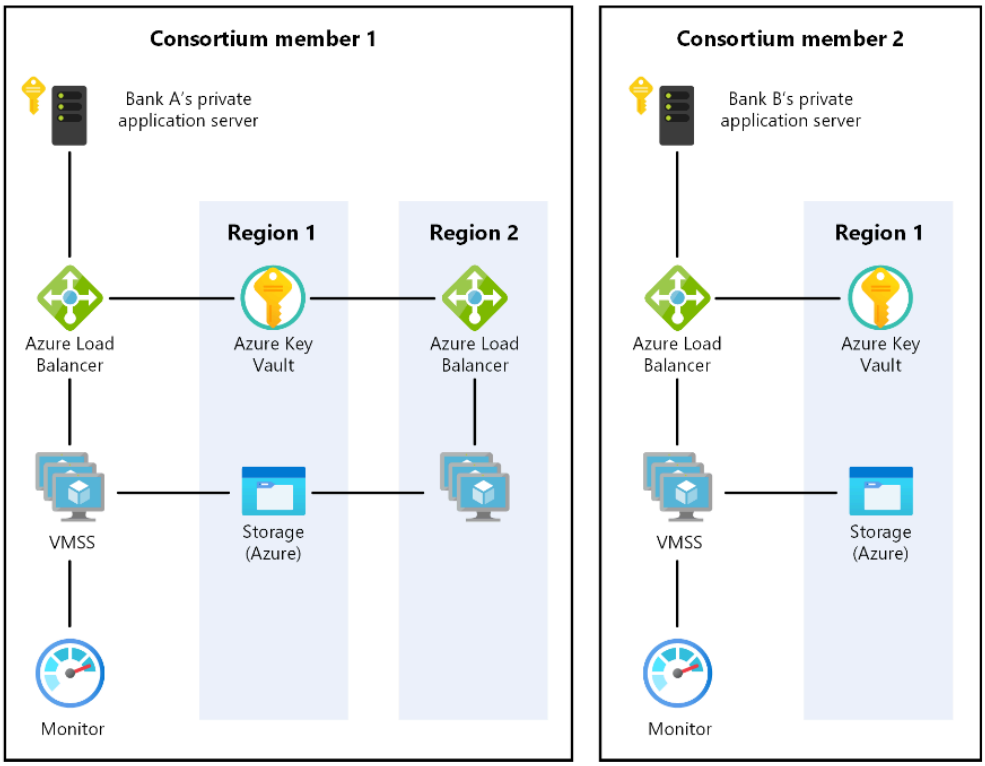
1. **Virtual machines (https://azure.microsoft.com/services/virtual-machines)**, within virtual machine scale sets, provide the on-demand compute facility to host the node for the DLT network.

2. **Azure Key Vault (https://azure.microsoft.com/services/key-vault/)** is used as the secure storage facility for private keys.

3. **Load Balancer (https://azure.microsoft.com/services/load-balancer/)** spreads communication requests to the VMs.

4. **Azure Storage (https://azure.microsoft.com/product-categories/storage)** hosts persistent network information and coordinates leasing.

5. Application Insights (part of Azure Monitor) can be used to provide insight into available nodes, transactions per minute, and consortium members.



Microsoft Azure

Reference: **https://learn.microsoft.com/en-us/azure/architecture/example-scenario/apps/decentralized-trust (https://learn.microsoft.com/en-us/azure/architecture/example-scenario/apps/decentralized-trust)**

Azure Functions

Azure Functions is a Platform as a Service that provides a serverless solution that allows you to write less code, maintain less infrastructure, and save on costs. Instead of worrying about deploying and maintaining servers, the cloud infrastructure provides all the up-to-date resources needed to keep your applications running. Here are some key details about Azure Functions:

- **Less code, maintain less infrastructure and save on costs:**

By using Azure Functions, you are only required to write the code necessary to perform a specific task or respond to a specific event. Azure manages the underlying infrastructure, such as servers and scaling, so you need to write less code and spend less time maintaining your application. In this manner, you can save money because you are only charged for the resources that are used during the execution of the function.

- **Azure Function supports two hosting/pricing models.**

a. Consumption –based plan:

This model allows you to pay for the execution time and resources consumed by your functions. According to the number of requests coming in, the platform automatically scales up or down the resources. It is a cost-effective solution for workloads with varying traffic patterns because you are only billed for the actual execution time of your functions.

b. Dedicated plan:

This model provides dedicated instances for your functions, which gives you greater control over their resources, performance, and scalability. It is suitable for workloads that require consistent performance or have specific networking or security requirements.

- **Designed for micro/nano – services which are small, decoupled components that perform a specific task or provide a focused functionality.**

For an overview of Azure Functions, please refer to the following resource.

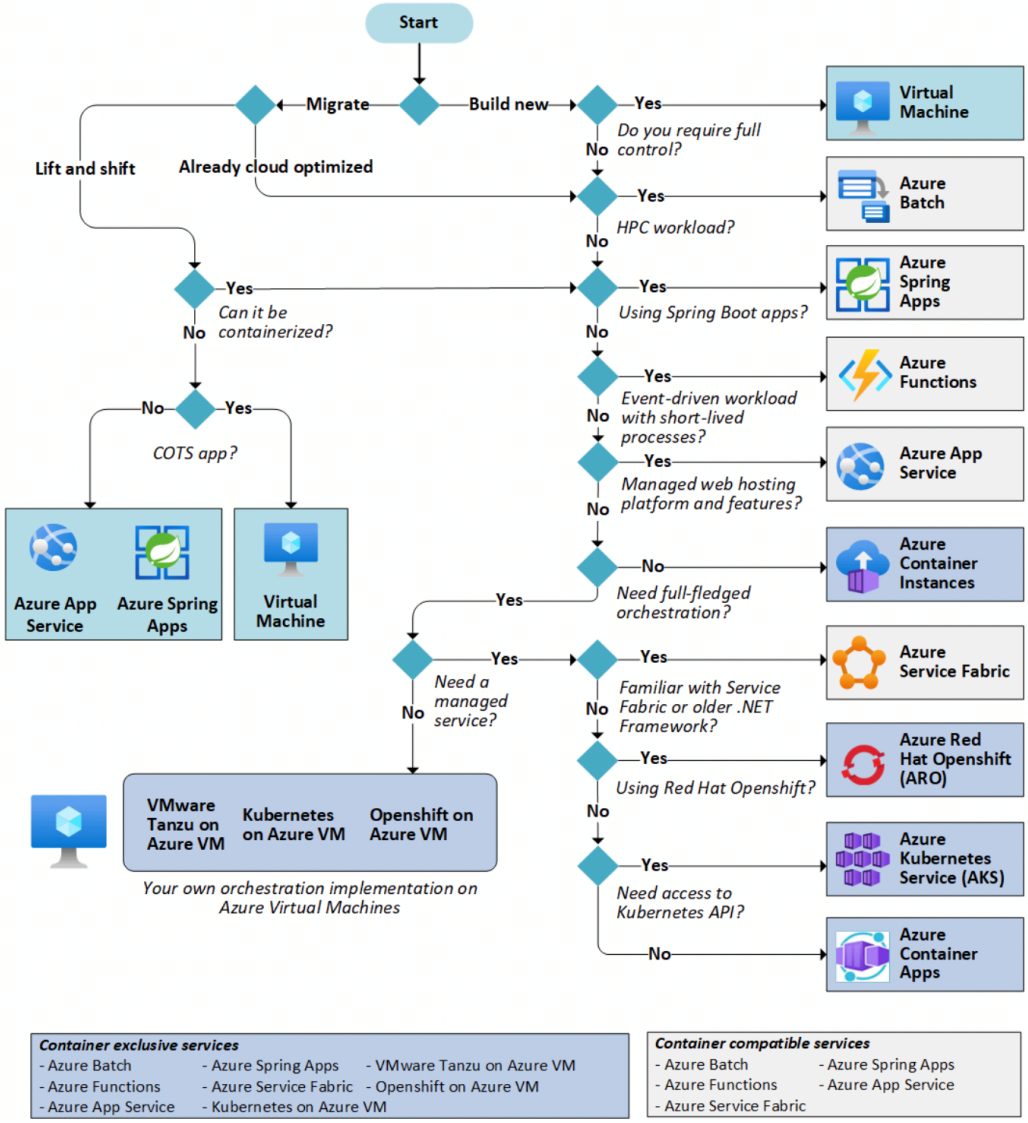
VIDEO LINK(HTTPS://YOUTU.BE/8-JZ5F_JYEQ)

Choosing an Azure Compute service

"Azure offers many ways to host your application code. The term compute refers to the hosting model for the resources that your application runs on. This section helps choose a compute service for your application".

-Reference: Microsoft, 2023.

The following flowchart will assist you in selecting a candidate compute service.



Reference: [https://learn.microsoft.com/en-us/azure/architecture/guide/technology-choices/compute-decision-tree (https://learn.microsoft.com/en-us/azure/architecture/guide/technology-choices/compute-decision-tree)](https://learn.microsoft.com/en-us/azure/architecture/guide/technology-choices/compute-decision-tree)

Azure Kubernetes Service (AKS)

For scenarios where you need full container orchestration, including service discovery across multiple containers, automatic scaling, and coordinated application upgrades, we recommend Azure Kubernetes Service (AKS). Here are some key features of Azure Kubernetes Service:

- **Managed container orchestration service:**

With AKS, developers can focus on building and deploying their applications by abstracting away the complexity of managing and operating Kubernetes clusters.

- **Open-source container orchestration platform** which has a large and active community of developers.

- **Platform as a Service (PaaS):**

AKS is a PaaS offering that provides a fully managed Kubernetes environment, including control plane and worker nodes.

- **Highly scalable and customizable:**

AKS allows users to easily scale their applications up or down based on demand.

- **Designed for highly scalable container deployments:**

AKS is designed for deploying and managing highly scalable containerized applications, making it ideal for applications with varying levels of demand.

In summary, Azure Kubernetes Service is a fully managed container orchestration service that provides a highly scalable and customizable environment for deploying and managing containerized applications using Kubernetes. AKS is designed for highly scalable container deployments, automates deployment and scaling, and integrates seamlessly with other Azure services. AKS is an ideal solution for developers who want to focus on building and deploying their applications, without worrying about the underlying infrastructure.

Azure Container Instance

Azure Container Instances is a platform as a service (PaaS) offered by Microsoft Azure that provides the simplest and fastest way to run containers in Azure, without having to manage any virtual machines and without having to adopt a higher-level service. This service can be used in a wide range of scenarios, including simple applications, task automation, and build jobs, which can be hosted in isolated containers on Azure.

Key Features

Here are some key features of Azure Container Instances:

Serverless containers:

In Azure Container Instance, serverless containers are supported, which means that users only pay for the resources they consume, without the need to provision or manage any infrastructure.

Designed for:

- Small and simple web apps/services
- Background jobs
- Scheduled scripts

Virtual Machine Scale Sets

Azure Virtual Machines Scale Set is an Infrastructure as a Service (IaaS) offering from Microsoft Azure, providing businesses high availability and application resiliency resources in the cloud.

Characteristics and Features

With Azure Virtual Machine Scale Sets, you can create and manage a group of load-balanced virtual machines. In response to demand or a defined schedule, the number of VM instances can automatically increase or decrease. Here are some key features and characteristics of Azure Virtual Machine Scale Sets:

- **Set of identical virtual machines:**

With Azure Virtual Machine Scale Sets, you can deploy and manage a set of identical virtual machines. As a result, it is easier to manage and maintain your virtual machine infrastructure, since all virtual machines can be updated and configured simultaneously.

- **Built-in auto-scaling features:**

Azure Virtual Machine Scale Sets includes built-in auto-scaling features that allow you to automatically scale the number of virtual machines up or down based on changes in demand. This ensures that your infrastructure can handle increased traffic or workloads without manual intervention.

- **Designed for manual and auto-scaled workloads:**

Azure Virtual Machine Scale Sets are designed to support both manual and auto-scaled workloads. This makes it suitable for a wide range of scenarios, including web services, batch processing, and other workloads that require scalability and flexibility.

For an overview of Azure Virtual Machine Scale Set and increase app availability and scalability, please refer to the following resource.

VIDEO LINK(HTTPS://WWW.YOUTUBE.COM/WATCH?V=LE2XJXYHNB8)

Containers

With Azure Container Apps, you can run microservices and containerized applications on a serverless platform, which is fully managed. Developers can deploy and manage containerized applications at scale with Azure Containers, a cloud-based container service. As a lightweight, portable, and efficient means of packaging and deploying software applications, containers are becoming more and more popular among developers of modern applications.

Characteristics and Features

Here are additional details on the key characteristics of Azure Containers:

- **Use host's operating system**

Containers in Azure are based on the host's operating system, which means they share the kernel with the host. Compared with traditional virtual machines, containers make better use of resources since they do not require the installation of a separate guest operating system.

- **Emulate OS (VM's emulate hardware)**

Containers emulate the operating system, rather than emulating hardware like virtual machines.

- **Lightweight (no OS)**
    - Development Effort
    - Maintenance
    - Compute & storage requirements

Due to the fact that containers only contain the components necessary to run the application, they are lightweight. Since containers do not require a full operating system to be installed, they can be deployed more quickly and require less storage space than virtual machines.

- **Respond quicker to demand changes**

Because containers can be scaled up or down based on workload demands, organizations are able to respond quickly to changes in demand.

- **Designed for almost any scenario**

Azure Containers are designed to handle almost any scenario, including web applications, microservices, and batch processing.

Azure Containers provide a lightweight, portable, and efficient method for packaging and deploying software applications. Compared to traditional virtual machines, they require less compute and storage resources, and can be scaled up and down quickly as required.

To learn more about Azure containers, please refer to the following resource for more detailed information regarding how to build and deliver Apps fast and scalable with Azure Container Apps.

VIDEO LINK(HTTPS://YOUTU.BE/SEYVXRPLIOE)

Azure Virtual Machines

A virtual machine is one of several types of on-demand, scalable computing resources that Azure provides. When you require more control over your computing environment than what is offered by other options, you usually choose a virtual machine. With Azure Virtual Machines, users are able to create and manage virtual machines in the cloud, while having full control over their operating system and software. Azure Virtual Machines is an Infrastructure as a Service (IaaS) offering from Microsoft Azure, providing businesses with scalable computing resources in the cloud. Using an Azure virtual machine allows you to take advantage of virtualization without purchasing and maintaining physical hardware. Nevertheless, it is necessary to maintain the virtual machine by performing tasks such as configuring, patching, and installing the software.

There are several ways in which Azure virtual machines can be utilized. The following are a few examples:

- Development and test
- Applications in the cloud
- Extended datacenter

Characteristics and Features

Your application can make use of as many virtual machines as is necessary to meet its requirements. The following are some of the key characteristics and features of Azure Virtual Machines:

Supports marketplace and custom images: There is a marketplace on Azure Virtual Machines where pre-configured images are available for popular operating systems and applications. For specific use cases, users may also create and upload custom images.

Best suited for:

- Custom software requiring custom system configuration
- Lift-and-shift scenarios

Can run any applications/scenarios

- Web apps and web services

- Databases
- Desktop applications
- Jumpboxes
- Gateways

Azure App Service

App service is an HTTP based service for hosting web applications, REST APIs, and mobile back ends. It supports multiple programming languages and containers. Depending on your preference, you can develop in .NET, .NET Core, Java, Ruby, Node.js, PHP, or Python. Containers such as Docker, Windows Server Containers, and Azure Functions are also supported.

Both Windows and Linux-based environments are capable of running and scaling applications with ease. Azure App Service is a fully managed platform as a service (PaaS), designed as an enterprise grade web application service. It connects with more than 50 connectors for enterprise systems (SAP, Salesforce), and internet services (such as Facebook).

App Service adds the power of Microsoft Azure to your application, such as security, load balancing, autoscaling, and automated management. Additionally, you can take advantage of its DevOps capabilities, such as continuous deployment from Azure DevOps, GitHub, Docker Hub, and other sources, package management, staging environments, custom domain, and TLS/SSL certificates.

With App Service, you pay for the Azure compute resources you use. The compute resources you use are determined by the App Service plan that you run your apps on.

For more information, see the Azure App Service plans overview here:

AZURE APP SERVICE PLAN OVERVIEW (HTTPS://LEARN.MICROSOFT.COM/EN-US/AZURE/APP-SERVICE/OVERVIEW-HOSTING-PLANS)

Developing and implementing Azure networking capabilities is an essential component of your cloud solution. You'll need to make networking design decisions to support your workloads and services properly. Several networking tools and capabilities are available through Azure. This section provides information about configuring network services in Azure, including virtual networks (VNets), subnets, and network security groups (NSGs). This module will demonstrate how to configure inbound and outbound connections using Azure Load Balancer, which is a fully managed service that distributes traffic across multiple resources, including virtual machines (VMs), virtual machine scale sets (VMSS), or application gateways. Additionally, you will learn about Azure Private Link, which is a secure and reliable method of accessing Azure services over a private network endpoint. By using Azure Private Link, you can connect to Azure services such as Azure Storage, Azure SQL Database, and Azure Cosmos DB with a private IP address rather than a public one. By doing so, you are able to keep your data within the network perimeter and prevent it from being exposed to the public Internet.

In this module, we will describe and design cloud network solutions for Azure. The following concepts will be covered:

1. Configure networking service
2. Inbound and outbound connections using Azure Load Balancer
3. Explore Azure Private Link

Learning Objectives

By the end of this module, you should be able to complete the following:

01 Illustrate Azure Virtual Networks

02 Explain Azure NSG

03 Clarify Azure VPN Gateway

04 Recall Azure Load Balancer

05 Identify Azure App Gateway

Azure Network Security Groups

Azure Network Security Groups (NSG), can filter network traffic between azure resources in Azure virtual network. A network security group contains security rules that allow or deny inbound network traffic to, or outbound network traffic from, several types of Azure resources. For each rule, you can specify source and destination, port, and protocol.

There are five-tuple information types, which are known as:

1. Source
2. Source port
3. Destination
4. Destination port; and
5. Protocol

They are used to evaluate and apply security rules. The priority and direction of two security rules cannot be the same. Existing connections are recorded in a flow record. Communication is allowed or denied based on the connection state of the flow record. It is possible to create a stateful network security group by utilizing the flow record. If you specify an outbound security rule to any address over port 80, for example, it's not necessary to specify an inbound security rule for the response to the outbound traffic. You only need to specify an inbound security rule if communication is initiated externally. The opposite is also true. If inbound traffic is allowed over a port, it's not necessary to specify an outbound security rule to respond to traffic over the port.

Existing connections may not be interrupted when you remove a security rule that allowed the connection. Modifying network security group rules will only affect new connections. When a new rule is created or an existing rule is updated in a network security group, it will only apply to new connections. Existing connections are not reevaluated with the new rules.

Application security groups enable you to configure network security as a natural extension of an application's structure, allowing you to group virtual machines and define network security policies based on those groups. You can reuse your security policy at scale without manual maintenance of explicit IP addresses. The platform handles the complexity of explicit IP addresses and multiple rule sets, allowing you to focus on your business logic. Application security groups have the following constraints:

1. There are limits to the number of application security groups you can have in a subscription, and other limits related to application security groups.
2. All network interfaces assigned to an application security group have to exist in the same virtual network that the first network interface assigned to the application security group is in.

3. If you specify an application security group as the source and destination in a security rule, the network interfaces in both application security groups must exist in the same virtual network.

Azure Virtual Network

Azure Virtual Networks (VNets) enable you to emulate physical network infrastructure within the cloud. A VNet is designed for isolation, segmentation, communication, filtering, and routing between resources, whether they are located on-premises or in the cloud. VNets are scoped for a single region; however, multiple virtual networks from different regions can be connected together using Virtual Network Peering.

The adoption of Azure and the cloud requires customers to have fast, private, and secure connections across regions and Azure Virtual Networks (VNets). Customer needs vary according to the type of workload. For example, in order to ensure data replication across geographies, you will require a high bandwidth, and a low latency connection. Depending on the customer's needs, Azure offers a variety of connectivity options for VNets, including peering and VPN connectivity options. VNet peering or VPN Gateway allows cross-VNet communication. Through VNet peering, two VNets within the same region can be connected. It allows resources within each VNet to communicate with each other as if they were on the same network.

A VNet is segmented into one or more subnets. Subnets enable you to segment the virtual network into one or more sub-networks and allocate a portion of the virtual network's address space to each subnet. You can then deploy Azure resources in a specific subnet. Just like in a traditional network, subnets allow you to segment your VNet address space into segments that are appropriate for the organization's internal network. This also improves address allocation efficiency. You can secure resources within subnets using Network Security Groups. Subnets are discrete solutions used for effective address allocation and Network filtering via Network Security Groups (NSG) or Application Security Groups (ASG).

In summary, Azure Virtual Networks provide you with the ability to emulate a physical network infrastructure within the cloud. It is designed for the purpose of isolating, segmenting, communicating, filtering, and routing between resources within a single geographic region. A VNet is comprised of one or more subnets that are used to efficiently allocate addresses, filter the network, and manage the resources within the VNet. A VPN Gateway and VNet peering enable cross-VNet communication within Azure or between on-premises networks and Azure VNets.

In this module, you learned various networking services provided by Azure and how to provision resources in Azure. A number of Azure networking sevices are described in some detail, such as Azure Virtual Networks, Azure Network Security Groups ,Azure VPN Gateway, Azure Load Balancer, and Azure Application Gateway
In the next lesson, you will learn to design and implement various cloud storage services and deep dive into a reference architecture for a concrete use case.

References

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Learning Activity 3.1: Reference architecture [Ungraded]



Let’s put what we learned in this module to practice, by creating your own architecture of Cloud network solution.

Scenario

The worldwide COVID-19 pandemic changed business, as the on-premises data platforms of many regulated industries were largely unable to cope with the increased workload resulting from the pandemic. These companies are now embracing cloud-based data platforms, but they're typically doing so in two extreme ways. Either with highly complex and detailed security requirements or with limited knowledge of cloud security capabilities and practices. This makes rapid deployment of services and solutions challenging because many security options can be disabled, overlooked, or simply ignored, leaving these companies open to regulatory action (like financial penalties) if left unchecked.

Dataflow

- Infrastructure and governance model:
  - The cloud ops team provisions, in a repeatable and consistent way, the data analysis environment. The team uses existing optimized security settings for regulated industries and an automated, parametrized process.
  - The network team typically integrates the environment with the enterprise network, ideally following the hub-and-spoke model with an enterprise firewall. This team also enables private links for endpoints and starts network traffic monitoring. We strongly recommend that you integrate Power BI with the virtual network, to use private traffic.
  - The cloud security team reviews the infrastructure through built-in or enterprise-specific Azure policies. This team reviews the security score of the environment in Azure Advisor or Azure Security Center.

Components

- Data Lake Store Gen2 (https://azure.microsoft.com/services/storage/data-lake-storage). Provides storage for business data, with snapshots, in raw and versioned format.

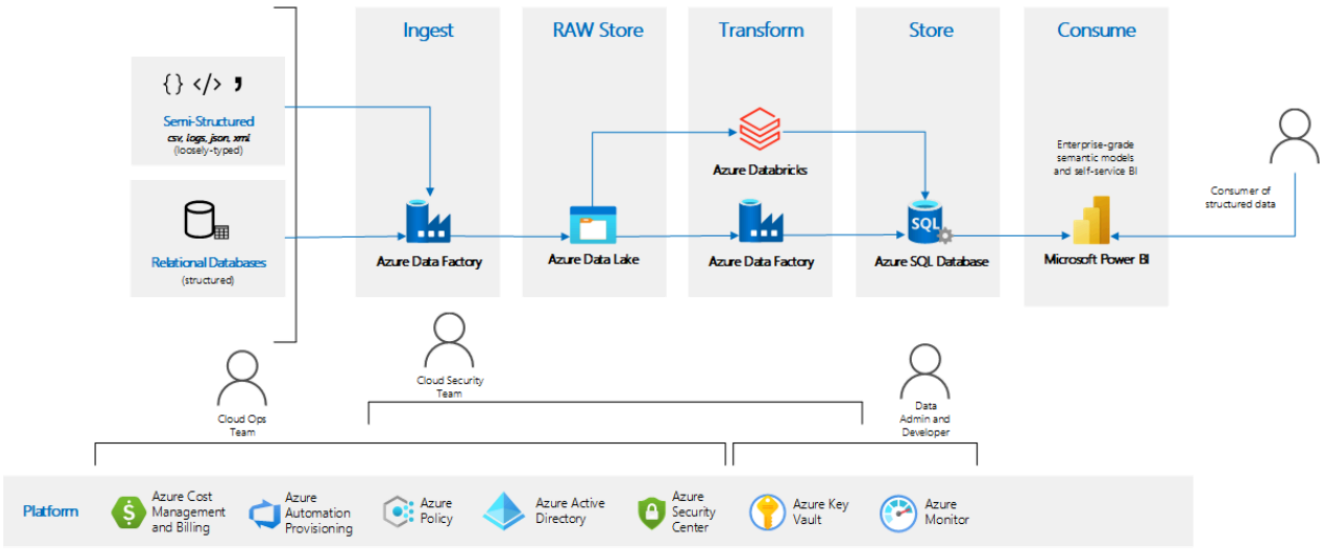
- SQL Database (https://azure.microsoft.com/products/azure-sql/database). A relational database service built for the cloud. In this architecture, it provides the data source that's consumed by Power BI clients.

- Data Factory (https://azure.microsoft.com/services/data-factory). Provides the data integration and transformation ETL/ELT layer.

- Azure Key Vault (https://azure.microsoft.com/services/key-vault). Used to store and protect keys and credentials in a secure place and monitor operations and access to it.

- Power BI, or Power BI Premium with VNet integration. (https://learn.microsoft.com/en-us/data-integration/vnet/use-data-gateways-sources-power-bi) (Optional.) Allows import or direct query datasets to connect to data services in an Azure virtual network without requiring an on-premises data gateway.

- Data Factory self-hosted integration runtime. (https://learn.microsoft.com/en-us/azure/data-factory/create-self-hosted-integration-runtime) (Optional.) Used to run activities between a cloud data store and a data store in a private network. It can also dispatch transform activities against compute resources in an on-premises network or an Azure virtual network



Reference: https://genesis-aka.net/information-technology/professional/2022/05/10/dataops-for-the-modern-data-warehouse/ (https://genesis-aka.net/information-technology/professional/2022/05/10/dataops-for-the-modern-data-warehouse/)

Tasks:

Using the information and diagram provided in the above scenario, take a moment to reflect on it.

- Now design your own architecture using the details from the scenario.

- Compare your own architecture with the suggested one

Azure Application Gateway

Azure Application Gateway is a web traffic load balancer that enables you to manage traffic to your web applications. It offers a variety of features to help enhance the performance, availability, and security of your web applications. Some of the key features of Azure Application Gateway include, but are not limited to:

Web application firewall

Web Application Firewall (WAF) is a service that provides centralized protection of your web applications from common exploits and vulnerabilities. Web applications are increasingly targets of malicious attacks that exploit common known vulnerabilities. Common among these exploits are SQL injection attacks, cross site scripting attacks, to name a few. Preventing such attacks in application code can be challenging and may require rigorous maintenance, patching and monitoring at many layers of the application topology. A centralized web application firewall helps make security management much simpler and gives better assurance to application administrators against threats or intrusions. A WAF solution can also react to a security threat faster by patching a known vulnerability at a central location versus securing each of the individual web applications. Existing application gateways can be converted to a Web Application Firewall enabled application gateway easily.

Redirection

A common scenario for many web applications is to support automatic HTTP to HTTPS redirection to ensure all communication between an application and its users occurs over an encrypted path. In the past, you may have used techniques such as dedicated pool creation whose sole purpose is to redirect requests it receives on HTTP to HTTPS. Application gateway supports the ability to redirect traffic on the Application Gateway. This simplifies application configuration, optimizes the resource usage, and supports new redirection scenarios, including global and path-based redirection. Application Gateway redirection support isn't limited to HTTP to HTTPS redirection alone. This is a generic redirection mechanism, so you can redirect from and to any port you define using rules. It also supports redirection to an external site as well.

Session Affinity

The cookie-based session affinity feature is useful when you want to keep a user session on the same server. Using gateway-managed cookies, the Application Gateway can direct subsequent traffic from a user session to the same server for processing. This is important in cases where session state is saved locally on the server for a user session.

URL Routing

URL Path Based Routing allows you to route traffic to backend server pools based on URL Paths of the request. One of the scenarios is to route requests for different content types to different pool.

SSL Termination

Application gateway supports Secure Socket Layers/ Transport Level Security (SSL/TLS) termination at the gateway, after which traffic typically flows unencrypted to the backend servers. This feature allows web servers to be unburdened from costly encryption and decryption overhead. But sometimes unencrypted communication to the servers isn't an acceptable option. This can be because of security requirements, compliance requirements, or the application may only accept a secure connection. For these applications, application gateway supports end to end SSL/TLS encryption.

Azure Load Balancer

Load balancing refers to even traffic distribution of load across a group of backend resources or servers. Azure Load Balancer operates at layer 4 of the Open Systems Interconnection (OSI) model. It's the single point of contact for clients. Load balancer distributes inbound flows that arrive at the load balancer's front end to backend pool instances. These flows are according to configured load-balancing rules and health probes. The backend pool instances can be Azure Virtual Machines or instances in a Virtual Machine Scale Set.

A public load balancer can provide outbound connections for virtual machines (VMs) inside your virtual network. These connections are accomplished by translating their private IP addresses to public IP addresses. Public Load Balancers are used to load balance internet traffic to your VMs. A public load balancer can provide outbound connections for virtual machines (VMs) inside your virtual network.

An internal or private load balancer is used where private IPs are needed at the frontend only. Internal load balancers are used to load balance traffic inside a virtual network These connections are accomplished by translating their private IP addresses to public IP addresses. Public Load Balancers are used to load balance internet traffic to your VMs.

Why use Azure Load Balancer?

With Azure Load Balancer, you can scale your applications and create highly available services. Load balancer supports both inbound and outbound scenarios. Load balancer provides low latency and high throughput and scales up to millions of flows for all TCP and UDP applications. To summarize, load balancer provides:

- High-availability and scalability scenarios
- Both TCP and UDP applications
- External and internal traffic

Azure VPN Gateway

Azure VPN Gateway is a service that uses a specific type of virtual network gateway to send encrypted traffic between azure virtual network and on-premise location over public internet. You can also use VPN Gateway to send encrypted traffic between Azure virtual networks over the Microsoft network. Multiple connections can be created to the same VPN gateway. When you create multiple connections, all VPN tunnels share the available gateway bandwidth.

VPN Gateway enables secure cross-regional communication of Azure virtual networks. With a VPN Gateway, resources in different regions can connect securely over a VPN connection to communicate with each other over VNets in different regions. A VPN gateway, however, can result in higher data transfer costs and more complex configurations of the network. Vnet peering vs VPN gateway should be chosen based on organization needs.

VNet Peering
VNet Peering provides a low latency, high bandwidth connection useful in scenarios such as cross-region data replication and database failover scenarios. Since traffic is completely private and remains on the Microsoft backbone, customers with strict data policies prefer to use VNet Peering as public internet is not involved. With no gateway in the path, there are no extra hops, ensuring low latency connections.

VPN Gateways
VPN Gateways provide a limited bandwidth connection and is useful in scenarios where encryption is needed, but bandwidth restrictions are tolerable. In these scenarios, customers are also not as latency-sensitive.

Key Terms and Concepts

Key Term or Concept	Definition
NSG	Network Security Groups
ASG	Application Security Groups
VNets	Virtual Networks
VMs	virtual machines

The Azure Storage platform is Microsoft's cloud storage solution for modern data storage scenarios. Azure Storage offers highly available, massively scalable, durable, and secure storage for a variety of data objects in the cloud. Azure Storage data objects are accessible from anywhere in the world over HTTP or HTTPS via a REST API. Azure Storage services offer durability, availability, security, scalability, manageability and accessibility as benefits for application developers and IT professionals.

In this module, we will describe and design a cloud storage solution with Azure focusing, more specifically on the following concepts:

1. Explore core storage services by azure
2. Storage account types in azure
3. Introduction to azure blob storage, azure files, azure queues, azure table storage

Learning Objectives

By the end of this module, you should be able to complete the following:

01 Recognize types of data

02 Clarify Blob storage including storage tiers

03 Explain File Storage

04 Illustrate Table Storage

05 Interpret Queue Storage

06 Identify Disk Storage

Azure table storage

Azure Table storage is a service that stores non-relational structured data (Storage for semi-structured data -NoSQL) in the cloud, providing a key/attribute store with a schema less design. Because Table storage is schema less, it's easy to adapt your data as the needs of your application evolve. Access to Table storage data is fast and cost-effective for many types of applications, and is typically lower in cost than traditional SQL for similar volumes of data.

You can use Table storage to store flexible datasets like user data for web applications, address books, device information, or other types of metadata your service requires. You can store any number of entities in a table, and a storage account may contain any number of tables, up to the capacity limit of the storage account.

Table storage

Azure Table storage stores large amounts of structured data. The service is a NoSQL datastore which accepts authenticated calls from inside and outside the Azure cloud. Azure tables are ideal for storing structured, non-relational data. Some characteristics of Table storage, include, but are not limited to:

- No need for foreign joins, foreign keys, relationships or strict schema.
- Designed for fast access.
- Providing key/attribute store with a schema less design.

Key Benefits

One of the key benefits of Azure Table Storage is the wide range of programming interfaces and SDKs available to developers. With Azure Table Storage, developers can choose the programming language and tools that are most comfortable for them, and it is easy to integrate them with their preferred programming language and tools. There are several programming interfaces supported, including .NET, Java, Python, Node.js, and others.

Another advantage of Azure Table Storage is that it provides access to table storage which is fast and cost-effective for many applications. It is capable of handling large volumes of data and provides fast read and write performance. Aside from its cost-effectiveness, Azure Table Storage offers a flexible pricing model that is based on usage.

Azure queue storage

Azure Queue Storage is a storage for small pieces of data (messages). You access messages from anywhere in the world via authenticated calls using HTTP or HTTPS. A queue message can be up to 64 KB in size. A queue may contain millions of messages, up to the total capacity limit of a storage account. Queues are designed for scalable asynchronous processing.
Queue Storage contains the following components:

- **URL format:**

Queues are addressable using the a URL.

- **Storage account**

All access to Azure Storage is done through a storage account.

- **Queue**

A queue contains a set of messages. The queue name **must** be all lowercase.

- **Message**

A message, in any format, of up to 64 KB. Before version 2017-07-29, the maximum time-to-live allowed is seven days. For version 2017-07-29 or later, the maximum time-to-live can be any positive number, or -1 indicating that the message doesn't expire. If this parameter is omitted, the default time-to-live is seven days.

Azure blob storage

Azure Blob Storage is Microsoft's object storage solution for the cloud. Blob Storage is optimized for storing massive amounts of unstructured data, such as text or binary data. Blob Storage is ideal for:

- Serving images or documents directly to a browser.
- Storing files for distributed access.
- Streaming video and audio.
- Storing data for backup and restore, disaster recovery, and archiving.
- Storing data for analysis by an on-premises or Azure-hosted service.

It is designed for storage of files of any kind (BLOB – Binary Large Object - File). Data stored in the cloud grows at an exponential pace, so to manage costs for your expanding storage needs, it can be helpful to organize your data based on how frequently it will be accessed and how long it will be retained. Azure storage offers different access tiers so that you can store your blob data in the most cost-effective manner based on how it's being used. Three storage tiers:

1. **Hot** – **frequently accessed data.** An online tier optimized for storing data that is accessed or modified frequently. The hot tier has the highest storage costs, but the lowest access costs.
2. **Cool** – **infrequently accessed data (lower availability, high durability).** An online tier optimized for storing data that is infrequently accessed or modified. Data in the cool tier should be stored for a minimum of 30 days. The cool tier has lower storage costs and higher access costs compared to the hot tier.
3. **Cold** - An online tier optimized for storing data that is infrequently accessed or modified. Data in the cold tier should be stored for a minimum of 90 days. The cold tier has lower storage costs and higher access costs compared to the cool tier.

NOTE: Archive - Rarely (if-ever) accessed data. An offline tier optimized for storing data that is rarely accessed, and that has flexible latency requirements, on the order of hours. Data in the archive tier should be stored for a minimum of 180 days.

When your data is stored in an online access tier (either hot, cool or cold), users can access it immediately. The hot tier is the best choice for data that is in active use. The cool or cold tier is ideal for data that is accessed less frequently, but that still must be available for reading and writing.

Usage scenarios for the hot tier include:

- Data that's in active use or data that you expect will require frequent reads and writes.
- Data that's staged for processing and eventual migration to the cool access tier.

Usage scenarios for the cool and cold access tiers include:

- Short-term data backup and disaster recovery.
- Older data sets that aren't used frequently, but are expected to be available for immediate access.
- Large data sets that need to be stored in a cost-effective way while other data is being gathered for processing.

Usage scenarios for the archive access tier include:

- Long-term backup, secondary backup, and archival datasets.
- Original (raw) data that must be preserved, even after it has been processed into final usable form.
- Compliance and archival data that needs to be stored for a long time and is hardly ever accessed.

Key Benefits

One of the key benefits of Azure Blob Storage is that it helps you create data lakes for analytics needs and provides storage to build cloud native and mobile apps. Azure Blob Storage can be accessed from anywhere, making it ideal for building cloud-native and mobile applications that require large amounts of data storage.

Types of data

The three main types of data are:

1. Structured data
2. Semi Structured data
3. Unstructured data

Various types of data have different formats, organization structures, and storage requirements. The importance of understanding the different types of data when designing a storage solution cannot be overstated. As each type of data has different storage and processing requirements, selecting the appropriate storage solution is dependent upon the type of data being stored. Over the course of this module, we will examine each of these data types a little further.

For now, as an excellent introduction into data types, please watch the following video titled "What is data? Different types of data? Structured | Semi-structured | Unstructured data".

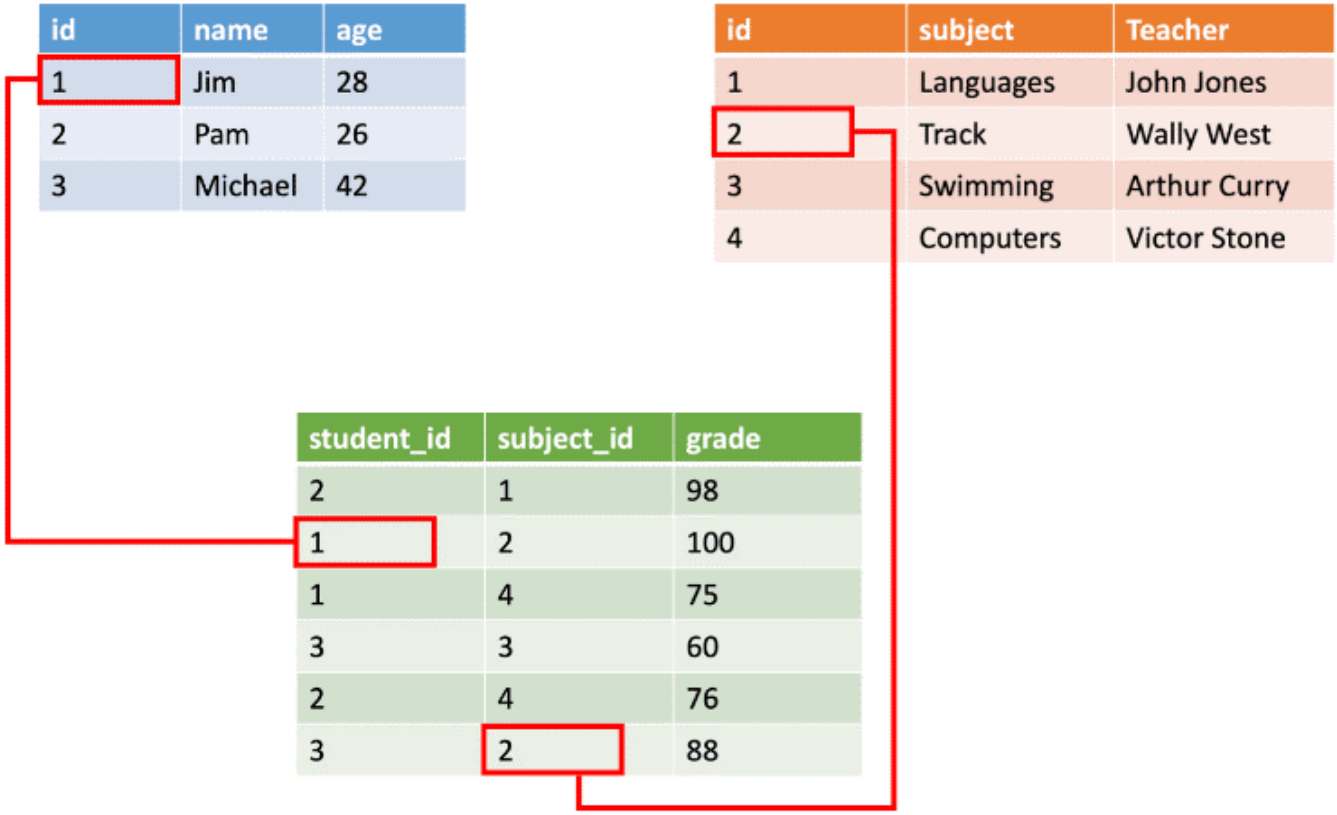
VIDEO LINK(HTTPS://WWW.YOUTUBE.COM/WATCH?V=CYLZIEXAATI)

Let's get started with structured data.

Structured Data

Structured data is generally tabular data that is represented by columns and rows in a database. Databases that hold tables in this form are called ***relational databases***. The mathematical term "*relation*" specify to a formed set of data held as a table. In structured data, all rows in the table has the same set of columns. SQL (Structured Query Language) programming language is used for structured data.

Structured Data looks something like this.



id	name	age
1	Jim	28
2	Pam	26
3	Michael	42

id	subject	Teacher
1	Languages	John Jones
2	Track	Wally West
3	Swimming	Arthur Curry
4	Computers	Victor Stone

student_id	subject_id	grade
2	1	98
1	2	100
1	4	75
3	3	60
2	4	76
3	2	88

Reference: **https://k21academy.com/microsoft-azure/dp-900/structured-data-vs-unstructured-data-vs-semi-structured-data/ (https://k21academy.com/microsoft-azure/dp-900/structured-data-vs-unstructured-data-vs-semi-structured-data/)**

Semi Structured Data

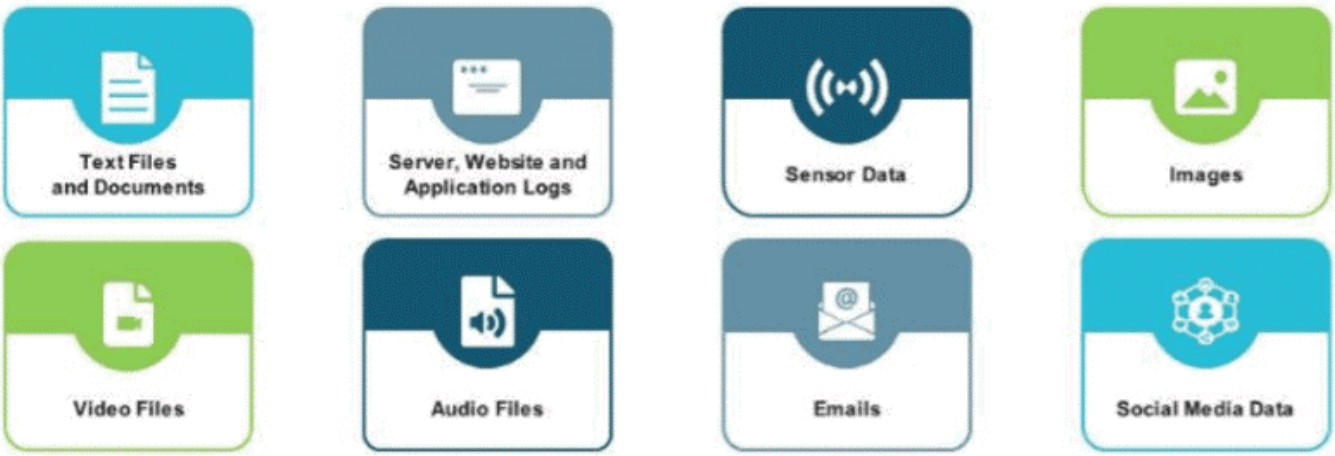
Semi-structured data is information that doesn't consist of Structured data (relational database) but still has some structure to it. Semi-structured data consist of documents held in JavaScript Object Notation (JSON) format. It also includes key-value stores and graph databases.
Semi Structured Data looks like something like this.

```
## Document 1 ##
{
  "customerID": "103248",
  "name":
  {
    "first": "AAA",
    "last": "BBB"
  },
  "address":
  {
    "street": "Main Street",
    "number": "101",
    "city": "Acity",
    "state": "NY"
  },
  "ccOnFile": "yes",
  "firstOrder": "02/28/2003"
}
```

Reference: **https://k21academy.com/microsoft-azure/dp-900/structured-data-vs-unstructured-data-vs-semi-structured-data/ (https://k21academy.com/microsoft-azure/dp-900/structured-data-vs-unstructured-data-vs-semi-structured-data/)**

Unstructured Data

Unstructured data is information that either does not organize in a pre-defined manner or not have a pre-defined data model. Unstructured information is a set of text-heavy but may contain data such as numbers, dates, and facts as well. Videos, audio, and binary data files might not have a specific structure. They're assigned to as unstructured data.



Reference: **https://k21academy.com/microsoft-azure/dp-900/structured-data-vs-unstructured-data-vs-semi-structured-data/ (https://k21academy.com/microsoft-azure/dp-900/structured-data-vs-unstructured-data-vs-semi-structured-data/)**

In this module, you learned various storage services provided by Azure and how to provision resources in Azure. A number of Azure storage services are described in some detail, including: Azure blob storage, Azure queue storage, Azure Table storage, Azure File storage, Azure storage account, and Azure Disk storage.

In the next module, you will learn to design and implement various cloud database services and deep dive into a reference architecture for a concrete use case.

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Learning Activity 4.1: Reference Architecture [Ungraded]



Just like we've don't in the previous modules, let's put what we learned here to practice, by creating your own architecture of Cloud Storage solution.

Scenario

Retail competition between companies is intense, and retailers are required to bring computational intelligence and smart insights into their commercial, marketing, and manufactory processes. While increasing and retaining customers is paramount for the success of a company, retailers must also align their inventory levels to customer demand to maximize operational efficiency and reduce waste and costs.

Dataflow

1. Azure Data Factory ingests related data into Azure Data Lake Storage. The sources of this data can be enterprise resource planning (ERP) systems, SAP, and Azure SQL. Additional sources might include weather and economic data.

2. Data Lake Storage stores raw data for further processing.

3. Mapping data flows in Azure Data Factory produces curated data and stores it in a relational format in an Azure SQL database. Additionally, in this use case, the SQL database stores intermediate results, other run information, and simulation metrics. Alternatively, Azure Machine Learning can read the data directly from the Data Lake service.

4. Use Azure Machine Learning to train the model by using data in Azure SQL Database and deploy the model and service to Kubernetes.

5. Install the Ray framework on the same Kubernetes cluster to parallelize the execution of the scoring script during inferencing. Each execution runs the demand-forecasting module for specified locations and products over a given forecast period. The forecasting results are read by the optimization module, which calculates the optimal inventory levels. Finally, the results are stored in Azure SQL Database.

6. Power Apps hosts a user interface for business users and analysts to collect parametric information, such as service level, product, and location. Users also use Power Apps to submit the collected parameters and to launch executions of the deployed machine learning module that is hosted in Kubernetes clusters.

7. Power BI ingests data from Azure SQL Database and allows users to analyze results and perform sensitive analysis. All Power BI dashboards are integrated into Power Apps to have a unified UI for calling the API, reading results, and performing downstream analysis.

Components

- **Data Lake Storage (https://azure.microsoft.com/services/storage/data-lake-storage)** is a scalable and secure data lake for high-performance analytics workloads. By using Data Lake Storage, you can manage petabytes of data with high throughput. Data Lake Storage can accommodate multiple, heterogeneous sources and data coming in structured, semi-structured, or unstructured formats.

- **Azure Data Factory (https://azure.microsoft.com/services/data-factory)** is a scalable and serverless service that provides a data-integration and transformation layer that works with various data stores.

- **Power BI (https://azure.microsoft.com/services/developer-tools/power-bi)** is a collection of software services, apps, and connectors that work together to turn your unrelated sources of data into coherent, visually immersive, and interactive insights.

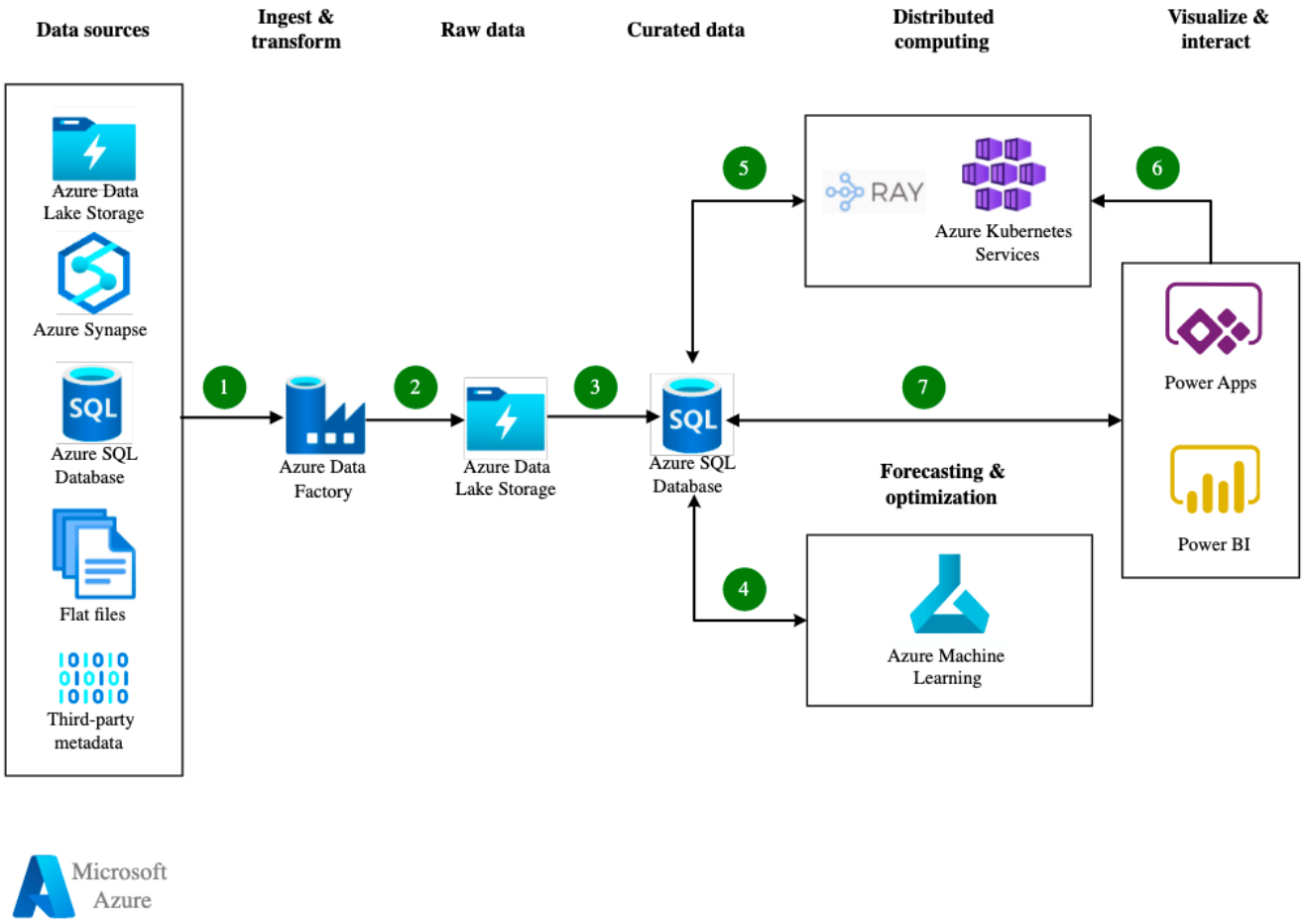
- **Azure Machine Learning (https://azure.microsoft.com/services/machine-learning)** is a cloud service for accelerating and managing the lifecycle of a machine learning project. Machine learning professionals, data scientists, and engineers can use Azure Machine Learning in their day-to-day workflows, to train and deploy models, and to manage Machine Learning Operations (MLOps).

- **Azure SQL Database (https://azure.microsoft.com/products/azure-sql/database)**: Azure SQL Database is a fully managed platform as a service (PaaS) that handles most of the database management functions, such as upgrading, patching, backups, and monitoring, without user involvement.

- **Azure Kubernetes Service (AKS) (https://azure.microsoft.com/services/kubernetes-service)**: AKS simplifies deploying a managed Kubernetes cluster in Azure by offloading the operational overhead to Azure. As a hosted Kubernetes service, Azure handles critical tasks like health monitoring and maintenance.

- **Azure Synapse Analytics (https://azure.microsoft.com/services/synapse-analytics)**: Azure Synapse Analytics is an enterprise analytics service that accelerates time to insight across data warehouses and big data systems. Azure Synapse Analytics brings together the following technologies:

- The best of the SQL technologies used in enterprise data warehousing
- Spark technologies used for big data
- Data Explorer for log and time-series analytics
- Pipelines for data integration and ETL/ELT
- Deep integration with other Azure services, such as Power BI, Azure Cosmos DB, and Machine Learning



Reference: [https://learn.microsoft.com/en-us/azure/architecture/example-scenario/analytics/optimize-inventory-forecast-demand (https://learn.microsoft.com/en-us/azure/architecture/example-scenario/analytics/optimize-inventory-forecast-demand)](https://learn.microsoft.com/en-us/azure/architecture/example-scenario/analytics/optimize-inventory-forecast-demand)

Tasks:

Using the information and diagram provided in the above scenario, take a moment to reflect on it.

- Now design your own architecture using the details from the scenario.
- Compare you own architecture with the suggested one

Azure disk storage

Azure managed disks are block-level storage volumes that are managed by Azure and used with Azure Virtual Machines. It emulates traditional disk storage in the cloud in an on-premises server but, virtualized. With managed disks, all you have to do is specify the disk size, the disk type, and provision the disk. Once you provision the disk, Azure handles the rest.

Key Benefits
One of the key benefits of Azure Disk Storage is that it provides persistent storage for virtual machines, allowing data to be stored even when the virtual machine is not running. This means that virtual machines can be stopped, started, or even deleted, without affecting the data stored on the disks. Azure Disk Storage also offers different sizes and types of disks to choose from, including solid-state drives (SSD) and hard disk drives (HDD).

Azure Disk Storage provides different performance tiers to meet various storage needs. The performance of your Azure managed disk is set when you create your disk, in the form of its performance tier. When you set the provisioned size of your disk, a performance tier is automatically selected. The performance tier determines the IOPS and throughput your managed disk has. The performance tier can be changed at deployment or afterwards, without changing the size of the disk and without downtime. Changing the performance tier allows you to prepare for and meet higher demand without using your disk's bursting capability. It can be more cost-effective to change your performance tier rather than rely on bursting, depending on how long the additional performance is necessary. This is ideal for events that temporarily require a consistently higher level of performance, like holiday shopping, performance testing, or running a training environment. To handle these events, you can switch a disk to a higher performance tier without downtime, for as long as you need the additional performance. You can then return to the original tier without downtime when the additional performance is no longer necessary.

Azure Disk Storage also offers the option of managed or unmanaged disks. A managed disk is fully managed by Azure and includes features such as automated backups, replications, and snapshots. Unmanaged disks, on the other hand, are managed by the user and require more manual configuration and management.

Azure storage account

An Azure storage account contains groups of services, which include:

- Blob storage
- Queue storage
- Table storage
- File storage

Azure Storage Account is used to store many data types such as: Files, Messages, and Semi-structured data. An additional advantage is its durability and 99.999999999% availability. This means data is protected against hardware failures, natural disasters, or other unexpected events. Additionally, Azure Storage Account ensures data availability even in the event of a data center outage by automatically replicating data between multiple data centers.

Furthermore, Azure Storage Accounts are highly scalable, enabling users to store and access petabytes of data. It also offers several types of storage accounts, which support different features and has its own pricing model.

Cost Effective
One of the most cost-effective cloud storage solutions is Azure Storage Account, which offers businesses and developers a variety of options for storing and managing their data. Among other cloud storage providers, Azure Storage Account provides the cheapest per GB storage solution.

Azure file storage

Azure Files offer fully managed file shares in the cloud that are accessible via the industry standard Server Message Block (SMB) protocol, Network File System (NFS) protocol, and Azure Files REST API.

Azure file is a storage for files accessed via shared drive protocols, and designed to extend on premise file shares or implement lift-shift scenarios.

For greater clarification on Azure File Storage and some of the common use cases for Azure Files, please watch the following videos:
(https://www.youtube.com/watch?v=H04e9AgbcSc)

VIDEO LINK(HTTPS://WWW.YOUTUBE.COM/WATCH?V=H04E9AGBCSC)

VIDEO LINK(HTTPS://WWW.YOUTUBE.COM/WATCH?V=9S5A1IJQFOQ)

Usefulness
Azure file shares can be used to:

Replace or supplement on premise file servers:
Azure Files can be used to replace or supplement traditional on premise file servers or network-attached storage (NAS) devices. Popular operating systems such as Windows, macOS, and Linux can directly mount Azure file shares wherever they are in the world. SMB Azure file shares can also be replicated with Azure File Sync to Windows servers, either on-premises or in the cloud, for performance and distributed caching of the data. With Azure Files AD Authentication, SMB Azure file shares can work with Active Directory Domain Services hosted on-premises for access control.

"Lift and shift" applications:
Azure Files makes it easy to "lift and shift" applications to the cloud that expect a file share to store file application or user data. Azure Files enables both the "classic" lift and shift scenario, where both the application and its data are moved to Azure, and the "hybrid" lift and shift scenario, where the application data is moved to Azure Files, and the application continues to run on-premises.

Simplify cloud development:
Azure Files can also be used to simplify new cloud development projects. For example:

Shared application settings:

A common pattern for distributed applications is to have configuration files in a centralized location where they can be accessed from many application instances. Application instances can load their configuration through the Azure Files REST API, and humans can access them by mounting the share locally.

Diagnostic share:

An Azure file share is a convenient place for cloud applications to write their logs, metrics, and crash dumps. Logs can be written by the application instances via the File REST API, and developers can access them by mounting the file share on their local machine. This enables great flexibility, as developers can embrace cloud development without having to abandon any existing tooling they know and love.

Dev/Test/Debug:

When developers or administrators are working on VMs in the cloud, they often need a set of tools or utilities. Copying such utilities and tools to each VM can be a time consuming exercise. By mounting an Azure file share locally on the VMs, a developer and administrator can quickly access their tools and utilities, no copying required.

Containerization:
Azure file shares can be used as persistent volumes for stateful containers. Containers deliver "build once, run anywhere" capabilities that enable developers to accelerate innovation. For the containers that access raw data at every start, a shared file system is required to allow these containers to access the file system no matter which instance they run on.

Key Terms and Concepts

Key Term or Concept	Definition
SMB	Server Message Block
NSF	Network File System
Durable and highly available	Redundancy ensures that your data is safe in the event of transient hardware failures. You can also opt to replicate data across data centers or geographical regions for additional protection from local catastrophe or natural disaster. Data replicated in this way remains highly available in the event of an unexpected outage.
Secure	All data written to an Azure storage account is encrypted by the service. Azure Storage provides you with fine-grained control over who has access to your data.
Scalable	Azure Storage is designed to be massively scalable to meet the data storage and performance needs of today's applications.
Managed	Azure handles hardware maintenance, updates, and critical issues for you.
Accessible	Data in Azure Storage is accessible from anywhere in the world over HTTP or HTTPS. Microsoft provides client libraries for Azure Storage in a variety of languages, including .NET, Java, Node.js, Python, PHP, Ruby, Go, and others, as well as a mature REST API. Azure Storage supports scripting in Azure PowerShell or Azure CLI. And the Azure portal and Azure Storage Explorer offers easy visual solutions for working with your data.

With Azure's fully managed databases, you can streamline your application development process. There are a number of database options available, including relational, NoSQL, and in-memory databases, covering both proprietary and open-source engines. By choosing fully managed databases, you can concentrate on building innovative applications while Azure handles the complexities on your behalf. Managed databases are capable of automating a number of tasks, including high availability, disaster recovery, backups, and data replication across different geographical locations. Thus, you are able to save time and resources, which ultimately results in a reduction in database management costs.

In this module, we will describe and design a cloud database solution with Azure, focusing, more specifically on the following concepts:

1. Implement Azure SQL databases
2. Introduction to Azure cosmos database and associated features and capabilities
3. Introduction to creating an app with cosmos dB

Learning Objectives

By the end of this module, you should be able to complete the following:

01 Explain Azure cosmos database

02 Interpret Azure SQL database

03 Clarify Azure SQL Managed Instance

04 Identify Azure database service for MySQL

05 Recognize Azure database service for PostgreSQL

Key Terms and Concepts

Key Term or Concept	Definition
DBaaS	Database as a service
SLA	Service Level Agreements

Azure Cosmos DB

Today's applications are required to be highly responsive and always online. To achieve low latency and high availability, instances of these applications need to be deployed in datacenters that are close to their users. Applications need to respond in real time to large changes in usage at peak hours, store ever increasing volumes of data, and make this data available to users in milliseconds.

Azure Cosmos DB is a fully managed NoSQL and relational database for modern app development. Azure Cosmos DB offers single-digit millisecond response times, automatic and instant scalability, along with guaranteed speed at any scale. Business continuity is assured with SLA-backed availability and enterprise-grade security. As a fully managed service, Azure Cosmos DB takes database administration off your hands with automatic management, updates and patching. It also handles capacity management with cost-effective serverless and automatic scaling options that respond to application needs to match capacity with demand.

The Azure Cosmos DB service is designed for:

- **Highly responsive** (real time) applications with super low latency responses < 10ms
- **Multi-regional** applications
- **Large volume** of data
- **Highly available** to users

Key benefit

1- **Guaranteed speed at any scale**

Gain unparalleled SLA-backed speed and throughput, fast global access, and instant elasticity. Real-time access with fast read and write latencies globally, and throughput and consistency all backed by SLAs. Multi-region writes and data distribution to any Azure region with just a button. Independently and elastically scale storage and throughput across any Azure region – even during unpredictable traffic bursts – for unlimited scale worldwide.

2- **Simplified application development:**

Build fast with open source APIs, multiple SDKs, schema-less data and no-ETL analytics over operational data.

- **Globally distributed**: NoSQL (semi-structured) database service. Build apps on API for NoSQL using the languages of your choice with SDKs for .NET, Java, Node.js and Python. Or your choice of drivers for any of the other database APIs.
- **Schema-less**: Azure Cosmos DB's schema-less service automatically indexes all your data, regardless of the data model, to deliver blazing fast queries.

- **Multiple APIs**: Choose from multiple database APIs including the native API for NoSQL, MongoDB, PostgreSQL, Apache Cassandra, Apache Gremlin, SQL and Table storage.

3- Mission-critical ready.

Guarantee business continuity, 99.999% availability, and enterprise-level security for every application.

4- Fully managed and cost-effective.

End-to-end database management, with serverless and automatic scaling matching your application and TCO needs.

For now, as an excellent introduction into Azure Cosmos DB, please watch the following video titled “What is Azure Cosmos DB? | Azure Cosmos DB Essentials Season 1”.

VIDEO LINK(HTTPS://WWW.YOUTUBE.COM/WATCH?V=JVGH64RVDXU)

Azure SQL Database

In today's data-driven world, driving digital transformation increasingly depends on our ability to manage massive amounts of data and harness its potential. But today's data estates are increasingly complex, with data hosted on-premises, in the cloud, or at the edge of the network. Developers who are building intelligent and immersive applications can find themselves constrained by limitations that can ultimately impact their experience. Limitations arising from incompatible platforms, inadequate data security, insufficient resources and price-performance barriers create complexity that can inhibit app modernization and development. Azure SQL is a family of managed, secure, and intelligent products that use the SQL Server database engine in the Azure cloud.

If you're new to Azure SQL, please check out the following video titled “What is Azure SQL?”.

WHAT IS AZURE SQL? (HTTPS://LEARN.MICROSOFT.COM/EN-US/SHOWS/AZURE-SQL-FOR-BEGINNERS/WHAT-IS-AZURE-SQL-3-OF-61)

Azure SQL Database is a relational database-as-a-service (DBaaS) hosted in Azure that falls into the industry category of Platform-as-a-Service (PaaS).

Azure SQL is:

- Best for modern cloud applications that want to use the latest stable SQL Server features and have time constraints in development and marketing.

- A fully managed SQL Server database engine, based on the latest stable Enterprise Edition of SQL Server. SQL Database has two deployment options built on standardized hardware and software that is owned, hosted, and maintained by Microsoft.

With SQL Server, you can use built-in features and functionality that requires extensive configuration (either on-premises or in an Azure virtual machine). When using SQL Database, you pay-as-you-go with options to scale up or out for greater power with no interruption. SQL Database has some additional features that are not available in SQL Server, such as built-in high availability, intelligence, and management.

Azure SQL Database offers the following deployment options:

1. As a single database with its own set of resources managed via a logical SQL server. A single database is similar to a contained database in SQL Server. This option is optimized for modern application development of new cloud-born applications. Hyperscale and serverless options are available.

2. An elastic pool, which is a collection of databases with a shared set of resources managed via a logical server. Single databases can be moved into and out of an elastic pool. This option is optimized for modern application development of new cloud-born applications using the

multi-tenant SaaS application pattern. Elastic pools provide a cost-effective solution for managing the performance of multiple databases that have variable usage patterns.

Key benefit

- Structured data service defined using schema and relationships.

Microsoft Azure SQL Database is a fully managed, relational database service. The purpose of this data service is to provide developers with a structured data service that is defined using schema and relationships, allowing them to work with data in a familiar and structured manner. The data can be organized into tables, each of which has a defined set of columns, and relationships can be established between tables to ensure consistency and accuracy.

- Rich Query Capabilities (SQL).

One of the key advantages of using Azure SQL Database is its rich query capabilities. Developers will be able to perform complex queries against their databases using SQL, a widely-used database query language. With SQL, developers can easily retrieve, insert, update, and delete data, as well as filter and aggregate data based on a variety of criteria. By doing this, developers are able to work efficiently and effectively when handling large datasets.

- High performance, reliable, fully managed and secure database for building applications.

Azure SQL Database also provides high performance and reliability. As a fully managed service, Azure SQL Database handles many aspects of database management, such as high availability, disaster recovery, backups, and data replication, allowing developers to focus on developing their applications. In addition to providing high levels of scalability and performance, Azure SQL Database is suitable for both small and large-scale applications. Another advantage of using Azure SQL Database is its security features. It provides advanced security features such as encryption at rest and in transit, auditing, threat detection, and dynamic data masking. By utilizing these features, developers can ensure that their data is secure and compliant with industry standards and regulations.

Azure SQL Managed Instance

Azure SQL Managed Instance is the intelligent, scalable cloud database service that combines the broadest SQL Server database engine compatibility with all the benefits of a fully managed and up-to-date platform as a service. SQL Managed Instance has near 100% compatibility with the latest SQL Server (Enterprise Edition) database engine, providing a native virtual network (VNet) implementation that addresses common security concerns, and a business model favorable for existing SQL Server customers. Azure SQL Managed Instance encompasses multiple capabilities that offer different ways organizations can optimize their workload costs without compromising quality and performance, allowing them to operate efficiently at a lower cost. Optimizing costs in SQL Managed Instance isn't a one-size-fits-all situation—there are several cost-saving benefits and capabilities to take advantage of depending on your unique business needs. By taking advantage of these benefits, or better yet, using two or more simultaneously, you can achieve significant cost savings while optimizing your business operations.

SQL Managed Instance allows existing SQL Server customers to lift and shift their on-premises applications to the cloud with minimal application and database changes. At the same time, SQL Managed Instance preserves all PaaS capabilities (automatic patching and version updates, automated backups, high availability) that drastically reduce management overhead and TCO.

Security Isolation

Additionally, Azure SQL Managed Instance provides secure data with native virtual network. SQL Managed Instance provides additional security isolation from other tenants on the Azure platform. Security isolation includes:

- Native virtual network implementation and connectivity to your on-premises environment using Azure ExpressRoute or VPN Gateway.
- In a default deployment, the SQL endpoint is exposed only through a private IP address, allowing safe connectivity from private Azure or hybrid networks.
- Single-tenant with dedicated underlying infrastructure (compute, storage).

To learn more about Azure SQL Managed Instance, please check out the following video titled "Azure SQL Managed Instance".

AZURE SQL MANAGED INSTANCE OVERVIEW (HTTPS://LEARN.MICROSOFT.COM/EN-US/SHOWS/AZURE-SQL-FOR-BEGINNERS/AZURE-SQL-MANAGED-INSTANCE-OVERVIEW-6-OF-61)

Azure Database for MySQL

Azure Database for MySQL is a relational database service in the Microsoft cloud based on the MySQL Community Edition (available under the GPLv2 license) database engine, versions 5.6 (retired), 5.7, and 8.0.

Azure Database for MySQL delivers:

- Zone redundant and same zone high availability.
- Maximum control with ability to select your scheduled maintenance window.
- Data protection using automatic backups and point-in-time-restore for up to 35 days.
- Automated patching and maintenance for underlying hardware, operating system and database engine to keep the service secure and up to date.
- Predictable performance, using inclusive pay-as-you-go pricing.
- Elastic scaling within seconds.
- Cost optimization controls with low cost burstable SKU and ability to stop/start server.
- Enterprise grade security, industry-leading compliance, and privacy to protect sensitive data at-rest and in-motion.
- Monitoring and automation to simplify management and monitoring for large-scale deployments.
- Industry-leading support experience.

These capabilities require almost no administration and are all provided at no extra cost. They allow you to focus on rapid app development and accelerating your time to market rather than allocating precious time and resources to managing virtual machines and infrastructure. In addition, you can continue to develop your application with the open-source tools and platform of your choice to deliver with the speed and efficiency your business demands, all without having to learn new skills.

Deployment Models

Azure Database for MySQL powered by the MySQL community edition is available in two deployment modes:

1. Flexible Server
2. Single Server

Let’s have a look at each of these servers in greater detail.

Flexible Server

Azure Database for MySQL Flexible Server is a fully managed production-ready database service designed for more granular control and flexibility over database management functions and configuration settings. The flexible server architecture allows users to opt for high availability within single availability zone and across multiple availability zones. Flexible servers provides better cost optimization controls with the ability to stop/start server and burstable compute tier, ideal for workloads that don't need full compute capacity continuously.

Single Server

Azure Database for MySQL single server is a fully managed database service for minimal customization. Handle most of database functions such as patching, backups, high availability, security and minimal user configuration and control. Single server is best suited for existing applications leveraging single server. For all new developments or migrations, Flexible Server would be the recommended deployment option.

Azure Database for PostgreSQL

Azure Database for PostgreSQL is a relational database service in the Microsoft cloud based on the PostgreSQL open source relational database.

Azure Database for PostgreSQL delivers:

- Built-in high availability.
- Data protection using automatic backups and point-in-time-restore for up to 35 days.
- Automated maintenance for underlying hardware, operating system and database engine to keep the service secure and up to date.
- Predictable performance, using inclusive pay-as-you-go pricing.
- Elastic scaling within seconds.
- Enterprise grade security and industry-leading compliance to protect sensitive data at-rest and in-motion.
- Monitoring and automation to simplify management and monitoring for large-scale deployments.
- Industry-leading support experience.

Deployment Models

Azure Database for PostgreSQL uses the same two deployment modes as MySQL, Single Server and Flexible Server.

Service Level Agreement

Azure's industry leading 99.99% availability service level agreement (SLA), powered by a global network of Microsoft-managed datacenters, helps keep your applications running 24/7. It offers SLA of up to 99.99 % uptime, highly available.

The solution offers numerous benefits, including lower total cost of ownership and enhanced operational efficiency. The reduction of infrastructure costs is one of the most significant advantages of Azure Database for PostgreSQL. Due to the fact that it is a fully managed service, organizations are no longer required to provision and manage their own infrastructure. This eliminates upfront hardware costs, reduces maintenance expenses, and frees up resources that would otherwise be allocated to infrastructure management. Moreover, Azure Database for PostgreSQL leverages open-source database management, allowing organizations to benefit from the cost savings associated with open-source software.

Furthermore, Azure Database for PostgreSQL incorporates performance optimization capabilities powered by artificial intelligence. A number of advanced algorithms and machine learning techniques are used to continuously monitor and analyze database workloads in order to provide intelligent recommendations for query optimization, index tuning, and performance enhancements. An organization can achieve higher efficiency and improve application response times by automating these performance optimizations, ultimately resulting in a better user experience. Azure Database for PostgreSQL places a high priority on security. The system is compliant with industry-leading security standards, including HIPAA, PCI, and DSS. Compliance with this standard ensures that sensitive data is protected and meets the strict regulatory requirements of various industries. For the protection of data and to prevent unauthorized access, Azure Database for PostgreSQL provides data encryption at rest and in transit, robust access controls, and integrated threat detection.

Azure Database for PostgreSQL also supports advanced features that facilitate the development of applications. This framework supports JSONB, which allows developers to store, query, and manipulate JSON data efficiently. Moreover, it supports geospatial data handling, allowing applications to handle location-based data. As well as offering rich indexing options, PostgreSQL offers dozens of extensions that extend its functionality, allowing developers to tailor their database to their particular application requirements.

Learning Activity 5.1: Reference Architecture [Ungraded]



Just like we've don't in the previous modules, let's put what we learned here to practice, by creating your own architecture of Cloud Database solution. This example scenario demonstrates a data pipeline that integrates large amounts of data from multiple sources into a unified analytics platform in Azure. This specific scenario is based on a sales and marketing solution, but the design patterns are relevant for many industries requiring advanced analytics of large datasets such as e-commerce, retail, and healthcare.

Scenario

The company needs a modern approach to analysis data, so that decisions are made using the right data at the right time. The company's goals include:

- Combining different kinds of data sources into a cloud-scale platform.
- Transforming source data into a common taxonomy and structure, to make the data consistent and easily compared.
- Loading data using a highly parallelized approach that can support thousands of incentive programs, without the high costs of deploying and maintaining on-premises infrastructure.
- Greatly reducing the time needed to gather and transform data, so you can focus on analyzing the data.

Dataflow

1. For each data source, any updates are exported periodically into a staging area in Azure Data Lake Storage.
2. Azure Data Factory incrementally loads the data from Azure Data Lake Storage into staging tables in Azure Synapse Analytics.
3. After loading a new batch of data into the warehouse

4. Business analysts use Microsoft Power BI to analyze warehoused data via the Analysis Services semantic model.

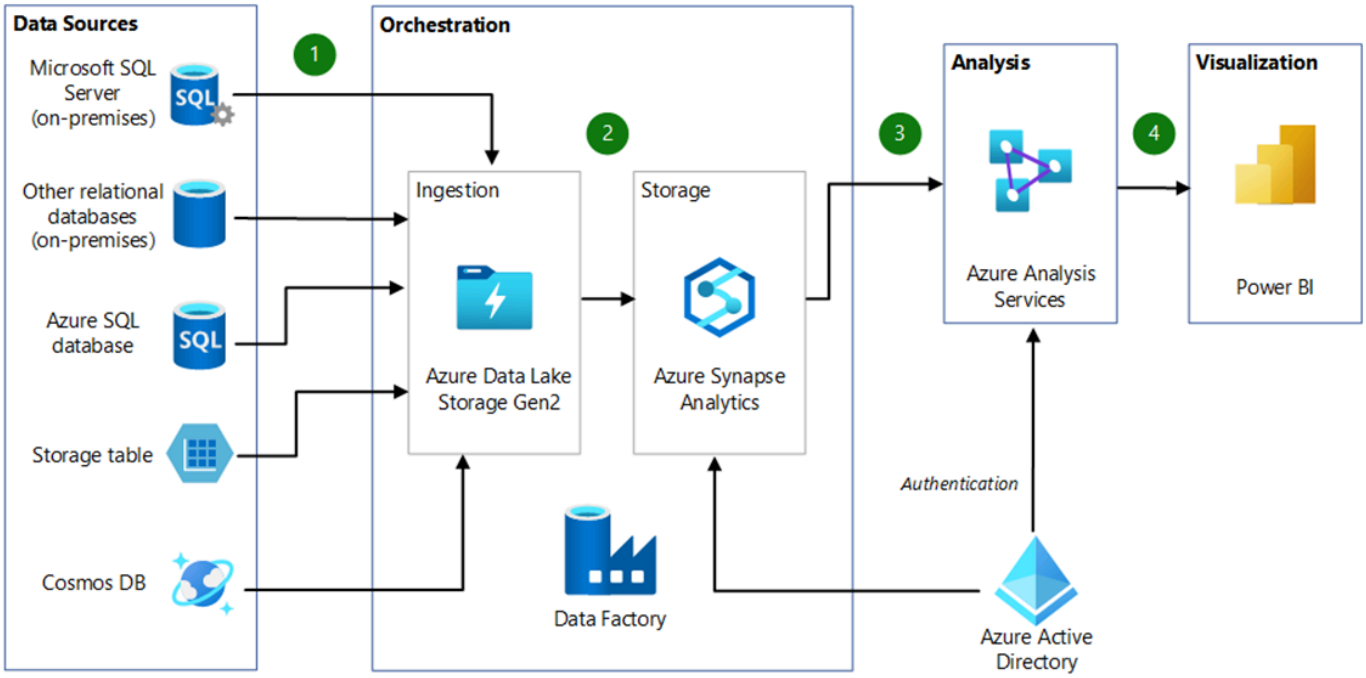
Components

The company has data sources on many different platforms:

- SQL Server on-premises
- Oracle on-premises
- Azure SQL Database
- Azure table storage
- Azure Cosmos DB

Data is loaded from these different data sources using several Azure components:

- **Azure Data Lake Storage (https://azure.microsoft.com/products/storage/data-lake-storage)** is used to stage source data before it's loaded into Azure Synapse.
- **Data Factory (https://azure.microsoft.com/products/data-factory)** orchestrates the transformation of staged data into a common structure in Azure Synapse. Data Factory **uses PolyBase when loading data into Azure Synapse (https://learn.microsoft.com/en-us/azure/data-factory/connector-azure-sql-data-warehouse#use-polybase-to-load-data-into-azure-sql-data-warehouse)** to maximize throughput.
- **Azure Synapse (https://azure.microsoft.com/products/synapse-analytics)** is a distributed system for storing and analyzing large datasets. Its use of massive parallel processing (MPP) makes it suitable for running high-performance analytics. Azure Synapse can use **PolyBase (https://learn.microsoft.com/en-us/sql/relational-databases/polybase/polybase-guide)** to rapidly load data from Azure Data Lake Storage.
- **Analysis Services (https://azure.microsoft.com/products/analysis-services)** provides a semantic model for your data. It can also increase system performance when analyzing your data.
- **Power BI (https://powerbi.microsoft.com/)** is a suite of business analytics tools to analyze data and share insights. Power BI can query a semantic model stored in Analysis Services, or it can query Azure Synapse directly.
- **Azure Active Directory (Azure AD) (https://azure.microsoft.com/products/active-directory)** authenticates users who connect to the Analysis Services server through Power BI. Data Factory can also use Azure AD to authenticate to Azure Synapse via a service principal or **Managed identity for Azure resources. (https://learn.microsoft.com/en-us/azure/active-directory/managed-identities-azure-resources/overview)**



Microsoft Azure

Reference: **https://learn.microsoft.com/en-us/azure/architecture/example-scenario/data/data-warehouse (https://learn.microsoft.com/en-us/azure/architecture/example-scenario/data/data-warehouse)**

Tasks:

Using the information and diagram provided in the above scenario, take a moment to reflect on it.

- Now design your own architecture using the details from the scenario.
- Compare you own architecture with the suggested one

The Cloud Adoption Framework is proven guidance that's designed to help you create and implement the business and technology strategies necessary for your organization to succeed in the cloud. It provides best practices, documentation, and tools in which cloud architects, IT professionals, and business decision makers need to successfully achieve their short- and long-term objectives. By using Cloud Adoption Framework best practices, organizations are better able to align their business and technical strategies to ensure success.

In this module, we will discuss various cloud architecture best practices, focusing, more specifically on the following concepts:

1. Decouple applications into collection of services
2. Optimize storage cost, availability, performance and data partitioning
3. Identify cloud architecture security best practices at every layer in the application

Learning Objectives

By the end of this module, you should be able to complete the following:

01 Explain the Cloud adoption framework

02 Recognize the correct cloud provider

03 Clarify Hybrid and multi-cloud options

04 Carryout the Automation to reduce costs

05 Implement security in every layer

06 Monitor all resources

07 Implement Elasticity

08 Decouple the cloud components

09 Implement Parallelization

10 Explore design principles for Azure Applications

Key Terms and Concepts

Key Term or Concept	Definition
MFA	Multi factor authentication

In this module, you learned various techniques and best practices to implement while designing cloud solutions. Cloud adoption framework was described in some details including: Strategy, Plan, Ready, Adopt, Manage, Govern phases. Choosing the right cloud provider, cost optimization, security, monitoring, and Elasticity were also discussed.

In the next module, you will expand on your knowledge of implementing the best architectural practices when designing cloud solutions.

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Elasticity

Elasticity is an ability of a system to scale-up during a sudden change in business needs, in order to handle the increased traffic load. Prior to implementing Elasticity, ensure that your deployment process is automated and build process and configuration are streamlined. There are three ways to implement Elasticity:

1. P**roactive Cyclic Scaling**: Periodic scaling (quarterly, monthly, weekly, daily)
2. **Proactive Event-based Scaling**: Scaling during the huge traffic surge (marketing campaigns, new product launch, festive season etc...)
3. **Auto-Scaling Based on Demand**: Scaling on the basis of system metrics

Monitor Resources

Azure Monitor provides comprehensive visibility to identify and diagnose problems in cloud based infrastructure. It’s important to monitor all resources like servers, applications, databases and networking components to ensure system runs smoothly and efficiently.

Monitoring tools include:

Activity log: Events that tracks operation for each azure resource.

Platform metrics: Describe some aspects of resource at a particular time.

Resource logs: These provide insights into operations performed by an azure resource.

Alerts: Alerts fired for the resource, it proactively notifies you when important conditions are found monitoring data and can use metrics or logs.

Security

We know that security is job one in the cloud and how important it is that you find accurate and timely information about Azure security. One of the best reasons to use Azure for your applications and services is to take advantage of its wide array of security tools and capabilities. These tools and capabilities help make it possible to create secure solutions on the secure Azure platform. Let's examine some of the major features of Azures security.

Shared responsibility model
Cloud provider and organization are equally responsible for security in cloud and that varies depending on the product type. As you consider and evaluate public cloud services, it's critical to understand the shared responsibility model and which security tasks are handled by the cloud provider and which tasks are handled by you. The workload responsibilities vary depending on whether the workload is hosted on Software as a Service (SaaS), Platform as a Service (PaaS), Infrastructure as a Service (IaaS), or in an on-premises datacenter.

Encrypting data at rest
Microsoft Azure includes tools to safeguard data according to your company's security and compliance needs. Encryption at Rest is a common security requirement. In Azure, organizations can encrypt data at rest without the risk or cost of a custom key management solution. Organizations have the option of letting Azure completely manage Encryption at Rest. Additionally, organizations have various options to closely manage encryption or encryption keys.

Encrypt data in transit
To help protect data in the cloud, you need to account for the possible states in which your data can occur, and what controls are available for that state. When data is being transferred between components, locations, or programs, it's in transit. An example is a transfer over the network, across a service bus (from on-premises to cloud and vice-versa, including hybrid connections such as ExpressRoute), or during an input/output process.

Identify all the sensitive information
Begin by conducting a thorough assessment of your Azure environment to identify and classify all sensitive information, such as personally identifiable information (PII), financial data, or intellectual property. This understanding is essential for implementing appropriate security controls.

Use key vault to store keys and secrets
Azure Key Vault provides a secure and centralized solution for managing cryptographic keys, certificates, and secrets. By leveraging Key Vault, you can safeguard sensitive information, control access to keys and secrets, and ensure secure storage and retrieval of cryptographic material.

Secure databases by using firewalls and access controls to limit access level of users, and devices
Implement network firewalls and access controls for your Azure databases to limit access to authorized users and devices. Configure firewall rules to allow access only from trusted networks or specific IP addresses. Additionally, enforce strong authentication mechanisms such as Azure Active Directory (Azure AD) for user authentication.

Log alerts

Implement robust logging and monitoring mechanisms in Azure to detect and respond to security incidents. Configure alerts and notifications based on predefined rules to proactively monitor for suspicious activities, unauthorized access attempts, or potential security breaches.

Use SAS enables to implement granular access controls

For Azure storage accounts, utilize Shared Access Signatures (SAS) to define granular permissions and time-limited access to resources. SAS tokens allow you to control access to storage resources, set expiration periods, and restrict specific operations, providing a fine-grained access control mechanism.

Enable MFA and complex passwords

Strengthen user authentication by enabling Multi-Factor Authentication (MFA) for Azure AD. MFA adds an extra layer of security by requiring users to provide additional verification factors, such as a phone app or SMS code, in addition to their password. Enforce the use of complex passwords and periodic password changes to enhance security.

Use Azure firewall and NSG to limit workloads

Leverage Azure Firewall and Network Security Groups (NSGs) to establish network security boundaries and restrict inbound and outbound traffic to your Azure workloads. Implement NSGs to define and enforce network traffic rules at the subnet or virtual machine level, and use Azure Firewall to centrally manage and control traffic across multiple virtual networks.

Define compliance objectives and standards

Identify your organization's compliance requirements and define specific objectives and standards for data security and privacy in Azure. Ensure adherence to relevant regulations and industry standards, such as GDPR, HIPAA, or ISO 27001. Implement necessary controls, conduct regular audits, and maintain documentation to demonstrate compliance with these standards.

Hybrid and multi-cloud options

Microsoft Azure provides all of the products and features required to help you build and operate your technology solutions in the cloud. We also understand that there are sound business reasons that may drive the necessity of using multiple private and public clouds. Let's quickly differentiate between hybrid and multi-cloud.

1. **Hybrid cloud**

- Helps to keep the data and applications on-premise while others moving to cloud
- Offers flexibility to choose workloads for environments

Furthermore, a hybrid cloud is a type of cloud computing that combines a private cloud (on-premises infrastructure), with a public cloud (computing services offered by third-party providers over the public internet). Hybrid clouds allow data and applications to consistently move between the two cloud environments. Many organizations choose a hybrid cloud strategy because of business requirements, such as meeting regulatory and data sovereignty requirements, maximizing on-premises technology investments, or addressing latency issues. The hybrid cloud is evolving to include edge workloads. Cloud-managed edge computing devices bring the computing power of the public cloud to the private cloud, which is closer to where the IoT devices reside, including data residing in applications, connected devices, and mobile consumer services. Reducing latency by moving workloads to the edge, allows devices to spend less time communicating with the cloud, and can operate reliably in extended offline periods. Expanded compute, storage, and service availability provides experience-driven resources closer to your customers.

2. **Multi-cloud**

- Involves two or more cloud providers
- Help you avoid vendor lock-in
- Advantage of the best features of each cloud platform

Additionally, Multi-cloud computing refers to the use of multiple cloud computing services from more than one cloud provider (including private and public clouds), in a heterogeneous environment. A multi-cloud strategy provides greater flexibility, and mitigates risk. Choose services from different cloud providers that are best suited for a specific task, or take advantage of services offered by a particular cloud provider in a specific location.

Choosing a Cloud Provider

When selecting a cloud provider, it is crucial to consider several factors, as they will ensure that they meet your specific needs. Based on these factors you can choose the right cloud provider best suited for you. Here are five important factors to consider when evaluating cloud providers:

1. **Security**
   Security is a top priority for any organization moving to the cloud. It is essential to assess the security measures implemented by the cloud provider to protect your data and applications. Look for providers that offer robust security features such as data encryption, network firewalls, identity and access management controls, intrusion detection and prevention systems, and regular security audits. Additionally, check if the provider complies with industry standards and regulations, such as ISO 27001 or SOC 2, to ensure they meet rigorous security requirements.

2. **Availability**
   High availability is critical to ensure that your applications and services are accessible and reliable. Evaluate the cloud provider's infrastructure and architecture to ensure they have redundant systems, data centers in multiple geographic regions, and failover mechanisms to minimize downtime. Look for providers that offer service level agreements (SLAs) guaranteeing a certain level of availability and uptime for your applications. Consider their track record for service outages and how they handle incidents to gauge their reliability.

3. **Performance of application**
   Application performance is crucial to provide a seamless user experience. Assess the cloud provider's network capabilities, including their network bandwidth, latency, and overall performance. Consider whether they offer content delivery networks (CDNs) or edge locations to optimize the delivery of content to end-users. Additionally, evaluate their infrastructure scalability and the ability to handle peak workloads to ensure that your applications can perform optimally under varying demands.

4. **Service Level Agreement (SLA) for uptime**
   The cloud provider's service level agreement (SLA) outlines the guaranteed uptime and availability of their services. Carefully review the SLA to understand the provider's commitments and any associated penalties for failing to meet those commitments. Look for providers that offer SLAs with high uptime guarantees, typically above 99.9%. Additionally, assess their track record in fulfilling SLAs and how they compensate customers for any service disruptions or breaches of agreed-upon uptime levels.

5. **Support from the providers**
   Technical support is essential for addressing any issues or concerns that may arise during your cloud deployment. Evaluate the level of support provided by the cloud provider, including the

availability of support channels (e.g., email, chat, and phone), response times, and expertise of their support team. Consider whether they offer 24/7 support and if they have dedicated account managers or customer success teams to assist you with your specific needs. It is also helpful to review customer reviews and feedback to gauge the provider's reputation for providing responsive and helpful support.

Manage

Cloud strategy requires solid planning, readiness, and adoption. But it's the ongoing operation of the digital assets that delivers tangible business outcomes. Without a plan for reliable, well-managed operations of the cloud solutions, those efforts will yield little value.
To prepare for cloud adoption lifecycle, the framework suggests the following procedure:

1. **Define the business commitments**
   Document supported workloads to establish operational commitments with the business and agreed upon cloud management investments for each workload.

2. **Establish the management baseline**
   Critically define classifications, cloud management tools and processes which are required to deliver minimum commitment to operations management.

3. **Expand the management baseline**
   Based on business commitments and operations decisions, make use of the included best practices to implement the required cloud management tooling.

4. **Advanced operations and design principles**
   Platforms or workloads requires a higher level of business commitment might require a deeper architecture review to deliver on resiliency and reliability commitments.

Cloud Governance

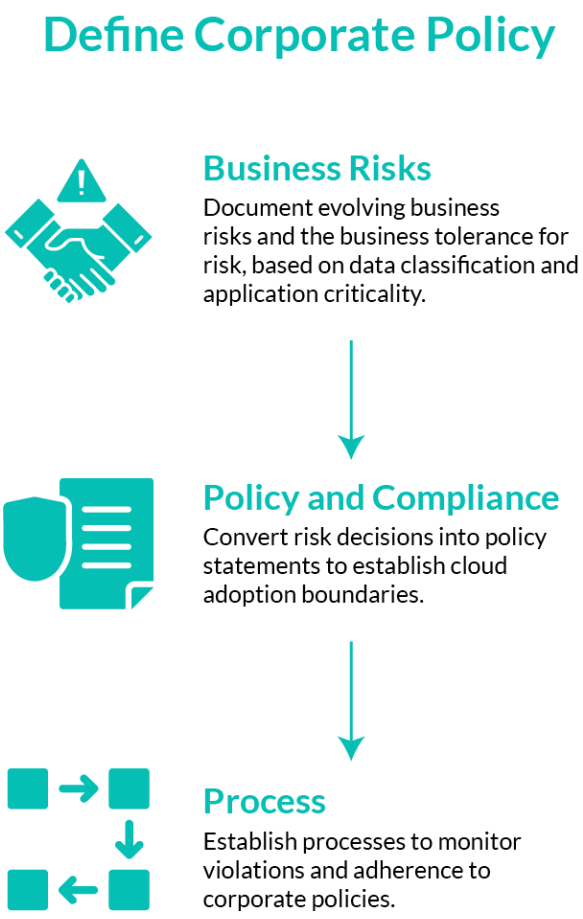
Cloud governance is an iterative process. For organizations with existing policies that govern on-premises IT environments, cloud governance should complement those policies. The level of corporate policy integration between on-premises and the cloud varies depending on cloud governance maturity and the nature of the digital estate in the cloud. As the cloud estate changes over time, so do cloud governance processes and policies. Use the following steps to help you start building your initial governance foundation:

1. **Establish your methodology**
   Establish a basic understanding of the methodology that drives cloud governance in the Cloud Adoption Framework to begin thinking through the end state solution.

2. **Use the governance benchmark**
   Assess your current state and future state to establish a vision for applying the framework.

3. **Establish the governance foundation**
   Begin your governance journey with a small, easily implemented set of governance tools. This initial governance foundation is called a minimum viable product (MVP).

4. **Improve your initial governance foundation**
   Throughout implementation of the cloud adoption plan, iteratively add governance controls to address tangible risks as you progress toward the end state.



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Additionally, it's important to note that Cloud Governance spans across the five disciplines:

- **Cost Management**

Evaluate & Monitor costs, limit IT spending, scale to meet need, create cost accountability.

- **Security Baseline**

Ensure compliance with IT Security requirements by applying a security baseline to all adoption efforts.

- **Resource Consistency**

Ensure consistency in resource configuration. Ensure practices for onboarding, recovery, and discoverability.

- **Identity Baseline**

Ensure the baseline for identity and access are enforced by consistently applying role definitions and assignments.

- **Deployment Acceleration**

Accelerate deployment through centralization, consistency, and standardization across deployment templates.

Ready

In preparing for a cloud adoption, a landing zone must be created, in order to host workloads which are planned to build in or migrate to the cloud. The following process helps to create a landing zone to support cloud adoption.

1. Cloud Operating Model

Teams establish models to define how technology would support a business. A cloud operating model is referred to as the collection of processes and procedures that defines how to operate technology on the cloud. IT operating model factors are essential for long-term technology operations. Some processes remain relevant when technology operations shift to the cloud. However, the processes might change in some ways. Current operating models focus on physical assets in physical locations, which are funded through capital expenditure cycles. These assets support workloads that the business needs to maintain operations. The mission of most operating models is to prioritize the workloads by investing in the stability of physical assets.

2. Azure landing zones

Landing zone is referred to as a multi-subscription azure environment that accounts to scale, security governance, networking and identity. It enables application migration, modernization and innovation at enterprise-scale in azure. This approach considers all platform resources that are required to support the customer's application portfolio and doesn't differentiate between infrastructure as a service or platform as a service.

3. Journey to architecture

Organizations iterate and mature the deployed technologies, processes and skills needed to progress forward to the destination. This iteration doesn't mean you arrive at the destination immediately, however. The journey takes time. The time varies depending on the size of the organization, the current technical footprint, and the skilling maturity within technical teams.

4. Azure landing zone design areas

Each Azure landing zone implementation option provides a deployment approach and defined design principles before choosing an implementation option.

Plan

Here, we are going to examine a few of the key steps necessary in helping you document your organization's technology strategy. These steps support cloud adoption efforts by prioritizing tasks. A Cloud adoption plan maps metrics and motivations defined in the strategy. The necessary steps are as follows:

1. **Digital Estate/Cloud Rationalization**: Conduct an inventory and rationalize your digital estate based on assumptions that align your organization's motivations and business outcomes.

2. **Initial organizational alignment:** Establish a plan for initial organizational alignment to support the adoption plan.

3. **Skills readiness plan:** Create a plan for addressing skills readiness gaps within your organization.

4. **Cloud adoption plan**: Develop a cloud adoption plan to manage change across skills, the digital estate, and your organization.

Let's examine these steps in greater detail.

Digital Estate/Cloud Rationalization

Process of evaluating assets to determine best way of migration or modernization each asset in cloud. There are 5 R's of rationalization:

1. Rehost

2. Refactor

3. Rearchitect

4. Rebuild

5. Replace

These cover most common options. It's easy to understand rationalization when you visualize the traditional process of rationalization as a complex decision tree. Each asset in the digital estate is fed through a process that results in one of five answers (the five Rs of rationalization). For small estates, this process works well. For larger estates, it's inefficient and can lead to significant delays.

Organizational alignment

Alignment of people who make the plan a reality. No plan is complete until you understand its people-related aspects. Proper organizational alignment takes time. It becomes essential to establish long-term organizational alignment, especially as cloud adoption scales across the

business and IT culture. Alignment is so important that there's an entire section dedicated to it in the Organize methodology of the Cloud Adoption Framework.

Skills readiness plan

Identify the skills that are necessary to support the responsibilities that come with digital transformation and check if your organization has the right skills. At each phase of the IT industry's history, the most notable changes have often been marked by changes in staff roles. One example is the transition from mainframe computing to client/server computing. The role of the computer operator during this transition has largely disappeared, replaced by the system administrator role. When virtualization arrived, the requirement for individuals working with physical servers was replaced with a need for virtualization specialists. Roles will likely change as institutions similarly shift to cloud computing. For example, datacenter specialists might be replaced with cloud administrators or cloud architects. In some cases, though IT job titles haven't changed, but the daily work of these roles has changed significantly.

Cloud adoption plan

An essential plan is required for the successful adoption of cloud as this helps a company to transition from traditional IT into modern, agile approaches. Cloud adoption plans start with a well-defined strategy. At a minimum, the strategy should outline the motivations, business outcomes, and business justifications for cloud adoption. Those positive returns are then balanced by the effort required to realize them. The effort starts with the digital estate (proposed or existing), which translates the strategy into more tangible workloads and assets. You can then map these tangible elements to technical work. From there, skilled people in a proper organizational structure can execute the technical work. The cloud adoption plan combines these topics into one plan that can be forecasted, budgeted, implemented, and managed by means of agile project-management practices.

Define Strategy

The cloud delivers fundamental technology benefits that can help your enterprise execute multiple business strategies. By using cloud-based approaches, you can:

1. Improve business agility
2. Reduce costs
3. Accelerate time to market
4. Enable expansion into new markets
5. Map cloud adoption strategy to cloud capabilities.

The following steps help you document your business strategy efficiently. This approach helps you drive adoption efforts that capture targeted business value in a cross-functional model. Then, you can map your cloud adoption strategy to specific cloud capabilities. You can also map business strategies to reach your expected state of transformation.

Steps

1. **Define and document your motivations behind cloud adoption:**
   Meet with key stakeholders and executives to document the motivations behind cloud adoption.
2. **Document business outcome:**
   Engage motivated stakeholders and executives to document specific business outcomes.
3. **Evaluate cost:**
   Learn how to use the cloud to make your IT cost structure more flexible. Then, build a business case to adopt the cloud.
4. **Understand technical aspects and capabilities while adopting cloud:**
   Discover the technical flexibility, efficiencies, and capabilities that help you build a business case to adopt the cloud.
5. **Cloud adoption strategy evaluator to access an overall cloud adoption strategy:**
   Help you build a business case to enable your cloud journey.

Cloud adoption framework

The Microsoft Cloud Adoption Framework for Azure is a full lifecycle framework that enables cloud architects, IT professionals, and business decision makers to achieve their cloud adoption goals. It provides best practices, documentation, and tools that help you create and implement business and technology strategies for the cloud. Following best practices for the Cloud Adoption Framework allows your organization to better align business and technical strategies and ensure success.

This framework helps you in understanding the organizations approach to accomplish a certain goal during their cloud adoption journey. The Cloud Adoption Framework brings together cloud adoption best practices from Microsoft employees, partners, and customers. The framework provides tools, guidance, and narratives. The tools it includes help you shape your technology, business, and people strategies to achieve the best business outcomes possible through your cloud adoption effort.

Your journey to the cloud can become more nuanced in complexity when you consider the implementation of larger, more complex cloud adoption scenarios that meet your unique business requirements. Choose the cloud adoption scenario that bests supports your strategy.

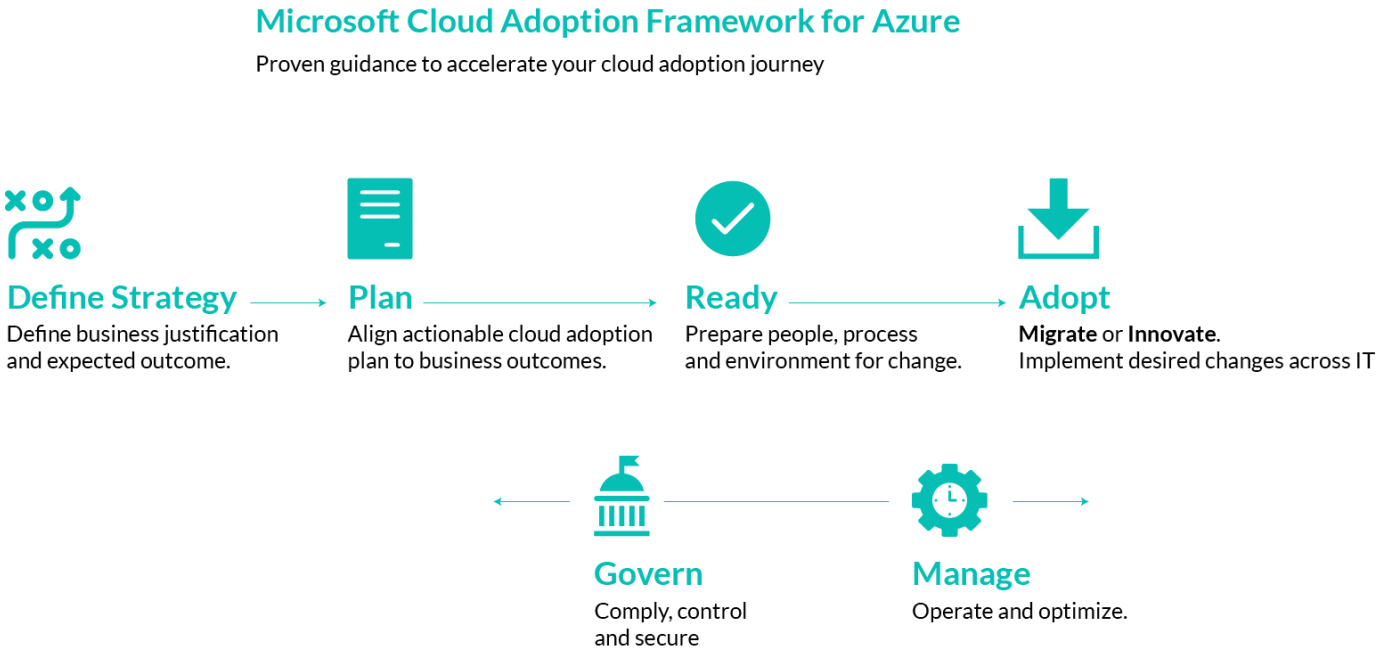
A team needs a cloud adoption plan to align their technical efforts with the organizations business strategy. Before adoption can begin, create a landing zone to host the workloads that you plan to build in or migrate to the cloud. The adopt methodology is where you start using the cloud and experiencing its benefits.

Cloud governance

Cloud governance is an iterative process. For organizations with existing policies that govern on-premises IT environments, cloud governance should complement those policies. The level of corporate policy integration between on-premises and the cloud, varies depending on cloud governance maturity and the nature of the digital estate in the cloud. As the cloud estate changes over time, so do cloud governance processes and policies.

Without a plan for reliable, well-managed operations of the cloud solutions, those efforts will yield little value. Cloud adoption plans convert the aspirational goals of a cloud adoption strategy into an actionable plan. Your collective cloud teams can use the cloud adoption plan to guide their technical efforts and align them with your organization's business strategy.

Just as cloud adoption is a journey, cloud security is also an ongoing journey of incremental progress and maturity, not a static destination. A journey without a target destination is just wandering. This approach might eventually lead to enlightenment, but business goals and constraints often require focusing on objectives and key results. The Secure methodology provides a vision of the complete end state, which guides the improvement of your security program over time.



Reference: **https://www.youtube.com/watch?v=pYxQp3jMv_U (https://www.youtube.com/watch?v=pYxQp3jMv_U)** Or **https://www.slideshare.net/abhishek22611/cloud-adoption-framework-overviewpartnerpptx (https://www.slideshare.net/abhishek22611/cloud-adoption-framework-overviewpartnerpptx)**

For now, as an excellent introduction into Azure Cloud adoption framework for Azure, please watch the following video titled “Microsoft Cloud Adoption Framework for Azure”.

VIDEO LINK(HTTPS://WWW.YOUTUBE.COM/WATCH?V=PYXQP3JMV_U)

Cost-optimization

Azure cost optimization is a crucial aspect of managing your cloud expenses efficiently. Here are further details:

1. **Aim for automated and end-to-end cost visibility**
   To effectively optimize costs, it's important to have a comprehensive view of your Azure spending. Leverage Azure's cost management and billing tools, such as Azure Cost Management + Billing, to gain insights into your usage patterns, resource costs, and trends. Set up automated reports and alerts to monitor and track your spending on an ongoing basis.

2. **Apply continuous azure cost monitoring and optimization**
   Cost optimization is an ongoing process. Regularly monitor your Azure usage and costs to identify areas where you can optimize spending. Utilize Azure's native monitoring and reporting tools to track resource usage, identify cost spikes, and detect any potential waste or inefficiencies.

3. **Automate azure cloud cost management**
   Automation can significantly aid in cost management. Leverage Azure Automation and Azure Logic Apps to automate routine tasks, such as resource provisioning and deprovisioning, based on predefined conditions or schedules. By automating these processes, you can ensure that resources are utilized efficiently and reduce unnecessary costs.

4. **Understand azure cloud costs**
   Gain a deep understanding of how Azure charges for different services and resources. Familiarize yourself with Azure's pricing models, including pay-as-you-go, reserved instances, and spot instances, to optimize cost based on your workload requirements. Explore Azure's pricing calculator and cost estimation tools to forecast expenses and make informed decisions about resource allocation.

5. **Resource tagging improves ability to see resources which are consuming azure budget**
   Implement resource tagging to categorize and label your Azure resources. By tagging resources, you can gain better visibility into how costs are distributed across different departments, projects, or applications. This enables you to identify areas of high spending and allocate costs accurately, improving cost allocation and budget management.

6. **Align VMs start/stop schedule with actual usage**
   Schedule the start and stop times of your virtual machines (VMs) based on actual usage patterns. By shutting down VMs during periods of inactivity, such as nights or weekends, you can reduce costs associated with idle resources. Leverage Azure Automation or scripting to automate the start and stop processes for VMs, ensuring optimal resource utilization.

7. **Shut down idle or unused resources**
   Identify and decommission any idle or unused resources in your Azure environment. This could include virtual machines, storage accounts, or databases that are no longer required. Regularly review your resource inventory and remove any unnecessary or underutilized resources to eliminate unnecessary costs.

8. **Rightsized underutilized resources**
   Analyze the performance and utilization of your resources, such as VMs, and ensure they are appropriately sized. Oversized resources can lead to unnecessary costs, while undersized resources can impact performance. Utilize Azure Advisor and monitoring tools to identify underutilized resources and size them accordingly.

9. **Elastic databases**
   Consider using Azure SQL Database elastic pools for managing multiple databases with variable workloads. Elastic pools allow you to share resources across multiple databases, optimizing resource utilization and reducing costs compared to individual databases with dedicated resources.

10. **Azure advisor uses data to recommend and improve cost savings, performance, reliability and security**
    Utilise Azure Advisor, an intelligent Azure service, to obtain recommendations and insights for optimizing costs. The Azure Advisor analyzes your usage patterns, resource configurations, and best practices to provide personalized recommendations for improving performance, reliability, and security.

11. **Azure policy to control costs**
    Azure Policy can be used to enforce governance and cost control measures. Using Azure Policy, you can define rules and policies to manage and restrict resource creation based on predefined conditions. As a result, accidental creation of costly resources can be avoided, and budgetary and compliance requirements can be adhered to.

Adopt

The adopt methodology is where you start using the cloud and experiencing its benefits. There are three common approaches for adopting the cloud:

1. Migration
2. Modernization; and
3. Innovation.

Migrate

- Exit the data center business
- Migrate infrastructure
- Use infrastructure as a service
- Improve

Modernize

- Reduce technical debt
- Modernize applications
- Use platform as a service
- Modernize databases

Innovate

- Reposition your business
- Develop technical solutions
- Use cloud native solutions
- Prioritize with data

Let’s examine each of these in greater detail.

Migrate

One of the key objectives of migration is to enable organizations to transition their infrastructure from the data center to the cloud. By migrating infrastructure to the cloud, businesses are able to benefit from the advantages of cloud computing and improve their operations. In this process, existing infrastructure components, such as servers, storage, and networking, are transferred from on-premises data centers to cloud-based platforms. Using Infrastructure as a Service (IaaS) offered by cloud providers is a popular option, as it allows organizations to offload the management and maintenance of their physical infrastructure. Migrating to the cloud enables organizations to achieve greater scalability, cost-effectiveness, and operational efficiency, which has a positive impact on their performance.

Modernize

Modernization focuses on reducing technical debt and upgrading applications to maximize their performance in a cloud environment. The modernization of applications can be accomplished through the adoption of modern software development practices, the use of Platform as a Service (PaaS) offerings, and the modernization of databases. By reducing technical debt, businesses can eliminate outdated technologies and practices that hinder their agility and scalability. Leveraging PaaS enables organizations to take advantage of cloud-native services and tools, allowing for faster development and deployment of applications. In addition, modernizing databases allows organizations to take advantage of advanced analytics capabilities provided by cloud platforms, as well as improve their data management capabilities.

Innovate

An important aspect of the adopted methodology is the innovation approach that emphasizes the repositioning of the business and the development of new technical solutions using cloud-native capabilities. Using the cloud, organizations can reimagine their business models, processes, and customer experiences. The task involves developing and deploying cloud-native solutions that utilize technologies such as artificial intelligence (AI), machine learning (ML), and the Internet of Things (IoT). Businesses can make informed decisions and drive innovation by prioritizing data analytics and insights. With cloud-native solutions, organizations can benefit from the scalability, agility, and advanced capabilities of the cloud, thereby increasing their competitiveness and differentiation.